Music School Workload Allocation & Resource Scheduling System

Milestone 1: Requirements Engineering & Object-Oriented Analysis Report

Course: INF2001 Introduction to Software Engineering

Institution: Singapore Institute of Technology

Team: P5-5

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1. Introduction

1.1 Team Composition & Contributions

Student Name	Student ID	Project Role	Primary Authored Sections & Responsibilities
Ryan Ke	2501911	Requirements Lead	Elicitation Protocol (§2.1), Functional Requirements Table 2.1, NFR Specifications Table 2.3, Teacher Use Cases UC-07 to UC-11 (§3.3.2).
Joseph Poon	2501079	Lead Architect	Executive Summary (§1.4), OOA Derivation (§4.1), Domain Class Diagrams Figures 4.1–4.2 (§4.2), Activity Diagrams Figures 3.2–3.4 (§3.4).
Wei Xiang	2500686	Business Logic Lead	Domain Business Rules, Manager Use Cases UC-01 to UC-06 (§3.3.1), Sequence Modeling SSD-01 to SSD-06 (§4.3), Legal Analysis (§2.2).
Tan Yan Qi	2500030	Scheduling Lead	Facilities/Studio Architecture, RTM Compilation Table 2.4 (§2.3), IT Admin Use Cases UC-12 to UC-15 (§3.3.3), Sequence Modeling SSD-07 to SSD-11 (§4.3).
Zayne Lee	2503844	Operations Lead	Reporting/Compliance Use Cases UC-16 to UC-17 (§3.3.4), Sequence Modeling SSD-12 to SSD-17 (§4.3), 3-Level WBS Table 5.1 (§5.1), Gantt Chart Figure 5.1 (§5.2).

1.2 Non-Plagiarism Declaration

Students are not allowed to copy assessment solutions from another person or other places. It is the students' responsibility to guarantee that their solutions are their own work. Meanwhile, the students must also ensure that their work is not accessible by others. Where such plagiarism is detected, both of the assessments involved will receive ZERO mark.

We hereby declare that:

- We fully understand and agree to the abovementioned policy.
- We did not copy any materials from others or from other places.
- We did not share our materials with others or upload to any other places for public access.
- We agree that we will not disclose any information or material of the team project to others or upload to any other places for public access.

- We agree that our project will receive Zero mark if any misalignment with the abovementioned policies is detected.

Declared by: Group P5-5
Team members: 1. Ryan Ke (2501911), 2. Joseph Poon (2501079), 3. Wei Xiang (2500686), 4. Tan Yan Qi (2500030), 5. Zayne Lee (2503844)
Date: 9 October 2026

1.3 AI Usage Declaration

In accordance with SIT academic policy, generative AI tools were used as assistive aids only, as listed below:

Document Section	Generative AI Tool	Specific Prompt Intent & Usage Scope	Human Governance, Review & Modifications
§2.2, §4.1.3 (Tables)	Claude Sonnet 5.5	Markdown table formatting, syntax linting, column alignment.	Replaced generic descriptions with concrete domain attributes; verified mathematical constraints.
§3.4, §4.2, §4.3 (Diagrams)	Claude Opus 5.5	PlantUML syntax verification, styling parameter checks.	Manually authored all workflow branches, lifelines, ECB dependencies, and guard logic.
§1.4, §2.1 (Narrative)	Claude Sonnet 5.5	Grammar checking and phrasing refinement.	Rewrote output to accurately reflect stakeholder interview evidence and statutory citations.
§5.2 (Project Schedule)	Claude Opus 5.5	PlantUML `@startgantt` syntax debugging.	Calculated task durations, realistic parallel concurrency, and verified M1 submission deadline (9 Oct 2026).

Engineering Integrity Confirmation: All core business logic—including the 4-hour continuous teaching limit with mandatory 1-hour rest, Studio 5 acoustic drum exclusivity, monthly 19th 23:59 cutoff window, 2-stage rejection lifecycle, and multi-constraint clash validation—was derived directly from stakeholder interviews and manually verified against project specifications. Automated scheduling algorithms were rejected to preserve human manager agency.

1.4 Executive Summary

Operational analysis of the client music academy revealed substantial friction stemming from its reliance on shared spreadsheets: an estimated 8 to 12 hours wasted weekly on manual timetable construction (as reported by operational management in interview MIN-01), recurrent studio double-bookings, unmonitored fatigue risks (teachers assigned >4 hours of continuous instruction without rest intervals), over-allocation beyond internal workload guidelines (>40 hours/week), and communication delays during teacher absences.

To resolve these operational deficits, Team P5-5 engineered a web-based decision-support system featuring real-time manpower intelligence, side-by-side 3-teacher allocation matrices displaying physical location and availability, Studio 5 drum kit exclusivity guards, and labor protection monitoring in accordance with Singapore's Employment Act 1968 (Republic of Singapore, 1968). Consistent with modern work-life balance frameworks (Guest, 2002; Lupu & Ruiz-Castro, 2021), sustainable workload distribution is structured as an ongoing governance process. This Milestone 1 report formalizes the software requirements, behavioral models, and object-oriented analysis architecture establishing the foundation for system development.

2. Requirement Engineering

2.1 Elicitation Methodology & Stakeholder Rationale

Requirements elicitation combined three complementary engineering techniques:

1. **Semi-Structured Stakeholder Interviews & Questionnaire Surveys:** Five semi-structured interviews, supplemented by an anonymous preliminary survey of music teachers on lesson clustering and fatigue recovery.
2. **Document & Artifact Domain Analysis:** Examined historical Excel timetables, room utilization logs, Singapore's Employment Act 1968 (Part IV: Hours of Work, Rest Days and Other Conditions of Service), and the Personal Data Protection Act (PDPA Section 24 Protection Obligation).
3. **Nominal & Exception Scenario Walkthroughs:** Traced monthly availability submissions (nominal pre-19th vs late post-19th exceptions), physical room conflict resolution, and emergency substitution workflows.

The five interviewees were two team members acting as internal representatives and three external representatives from other project teams:

Session ID	Stakeholder Role	Representative Name & Role	Primary Operational Concerns	Derived System Responsibilities
MIN-01-MGR	School Co-Manager	Marcus Tan *(Operations Co-Manager, Internal)*	Spreadsheet collisions, manual tallies, unmonitored overtime.	Workload dashboard, 3-teacher allocation matrix, conflict validation engine.
MIN-02-TCH	Music Teacher	Elena Lim *(Senior Instructor, Internal)*	Lost preferences, fatigue from unbuffered blocks, schedule volatility.	1-month availability grid (editable to 19th), preferences, 4h teaching block ceiling.
MIN-03-IT	IT Administrator	Alex Wong *(Systems Administrator, External)*	Excel corruption, lack of RBAC, untracked maintenance blackouts.	User lifecycle management, studio/instrument configuration, maintenance scheduling.

MIN-04-STU	Music Student	Chloe Teo *(Student Representative, External)*	Rescheduling communication lag (24–48h), teacher continuity.	Domain attributes in Lesson; priority queue handling for rescheduled lessons.
MIN-05-MGT-EXT	School Management	David Koh *(Academy Director, External)*	Perishable studio capacity, teacher turnover, Employment Act compliance.	Studio utilization analytics, workload equity reports, 24-month audit trail.

2.2 Software Requirement Specifications (SRS)

Requirements are prioritized using the MoSCoW framework (**M**ust have, **S**hould have, **C**ould have, **W**on't have).

Table 2.1: Master Functional Requirements Specification (FR-01 to FR-19)

ID	P	Functional Requirement Specification	Mapped UC
FR-01	**M**	The system shall calculate and display the running weekly and monthly allocated hours (`runningWeeklyHours`, `runningMonthlyHours`) for each teacher upon lesson confirmation.	`UC-01`
FR-02	**M**	The system shall sort and display the top 3 teachers with the lowest monthly allocated hours on the manager dashboard and flag teachers whose weekly hours exceed 40.0 hours.	`UC-01`
FR-03	**M**	The system shall allow managers to manually allocate unassigned 30-minute lesson slots to teachers and studios for a target week.	`UC-02`
FR-04	**M**	The system shall display up to 3 candidate teachers side-by-side during allocation, showing current workload, preferences, and physical studio location for a specified date.	`UC-02`
FR-05	**S**	The system shall rank eligible candidate teachers by remaining weekly hours below the 40.0-hour threshold (`40.0 - runningWeeklyHours`) and preferred teaching day match.	`UC-02`
FR-06	**M**	The system shall allow teachers to submit and edit recurring 30-minute availability slots up to 1 month ahead until 23:59 on the 19th of each month.	`UC-07`

FR-07	**M**	The system shall allow teachers to specify preferred teaching days, preferred studio ID, and consecutive lesson block preferences (up to 4.0 continuous hours followed by a minimum 1.0-hour rest break) within their submitted availability window.	UC-08
FR-08	**M**	The system shall provide teachers with a personal weekly schedule displaying confirmed lessons, running monthly hours, and masked student contact details.	UC-09
FR-09	**M**	The system shall display a mandatory consultation warning modal upon teacher job rejection and route confirmed rejection requests to the manager exception queue for re-allocation.	UC-03, UC-10
FR-10	**S**	The system shall flag post-19th availability changes as `Late - Pending Manager Review` and route them to the manager exception queue.	UC-04
FR-11	**C**	The system shall allow teachers to submit bilateral lesson swap proposals subject to instrument qualification verification, clash validation, and manager sign-off.	UC-11
FR-12	**S**	The system shall allow IT administrators to schedule maintenance blackout windows for studios and suppress blacked-out studios from lesson allocation.	UC-15
FR-13	**S**	The system shall allow managers to reassign scheduled lessons of an absent teacher to qualified and available substitute teachers.	UC-06
FR-14	**M**	The system shall allow IT administrators to provision user accounts and perform logical account deactivations for Teachers, Managers, and IT Administrators.	UC-12, UC-13
FR-15	**M**	The system shall manage studio configuration parameters for Studios 1 to 5 and restrict drum lesson allocations exclusively to Studio 5.	UC-14
FR-16	**M**	The system shall allow managers to transition draft weekly schedules to published status and notify assigned teachers.	UC-05
FR-17	**S**	The system shall calculate booked versus available studio hours across the 86 weekly operating hours and export studio utilization reports.	UC-16
FR-18	**M**	The system shall track teacher weekly working hours against the 44.0-hour statutory limit under Employment Act Part IV and flag authorized overtime between 40.5 and 44.0 hours for review.	UC-17

FR-19	**M**	The system shall append an immutable audit log record for every lesson allocation, managerial override, status transition, and administrative update, and allow managers to query and inspect the audit trail.	UC-17

Table 2.2: Won't-have Requirements (Out of Scope for M1)

Item	Justification & Architectural Boundary
Student self-service booking	Lessons are managed via administrative decision-support; direct student booking is an out-of-scope extension.
Student self-service account login	Student data is maintained administratively as entities/attributes; no external student portal in M1.
Online fee payment processing	Financial billing is handled by external accounting systems, not workload scheduling.
Payroll wage disbursement	FR-18 monitors compliance hours and overtime multipliers; monetary payroll processing is strictly out of scope.
Automated AI roster optimization	Excluded per project baseline agreement; human managers retain full agency over allocations.
Native mobile application	System is engineered as a responsive web application (viewports 375px–1920px), eliminating app store overhead.

Table 2.3: Quantifiable Non-Functional Requirements (NFR)

NFR ID	Category	Non-Functional Requirement Specification	Statutory / Business Justification
NFR-USE-01	Usability	The system shall render all user interfaces across mobile and desktop web browsers without horizontal scrolling.	Enables mobile schedule review for teachers and desktop allocation for managers.
NFR-USE-02	Usability	The system shall display teacher workload status using three visual tiers: Green (<35.0h), Amber (35.0h–40.0h), and Red (>40.0h) paired with text status labels.	Prevents over-allocation and ensures clarity for color-deficient users.
NFR-USE-03	Usability	The system shall provide interactive controls with a minimum target size of 44 by 44 CSS pixels on mobile views.	Supports accurate touch interaction for instructors between sessions.

NFR-PER-01	Performance	The system shall complete initial page rendering within 1.5 seconds under a concurrent load of 20 active users.	Ensures responsive management operations during monthly scheduling cycles.
NFR-PER-02	Performance	The system shall execute multi-constraint clash validation and return eligibility results within 300 milliseconds of slot selection.	Delivers immediate decision feedback during interactive lesson allocation.
NFR-PER-03	Performance	The system shall propagate studio configuration updates to the scheduling validation engine within 500 milliseconds of persistence.	Guarantees immediate constraint enforcement after administrative updates.
NFR-REL-01	Reliability	The system shall maintain at least 99.5% uptime during the monthly scheduling window from the 19th to the 24th of each calendar month.	Prevents administrative bottlenecks during monthly roster generation.
NFR-REL-02	Reliability	The system shall execute automated daily database backups supporting a Recovery Point Objective (RPO) of 24 hours and a Recovery Time Objective (RTO) of 2 hours.	Protects academy scheduling history and compliance audit data from data loss.
NFR-REL-03	Reliability	The system shall execute account and lesson removals exclusively via logical status flags without physical record deletion.	Maintains unbroken historical auditability and referential integrity.
NFR-REL-04	Reliability	The system shall cache unsubmitted availability form drafts in browser storage for up to 5 minutes during network disconnections.	Protects teachers against losing availability edits during transient Wi-Fi dropouts.
NFR-SEC-01	Security	The system shall encrypt personal data at rest and in transit and mask student phone numbers and contact details in all teacher views.	Complies with the Singapore Personal Data Protection Act (PDPA Section 24).
NFR-SEC-02	Security	The system shall enforce role-based access control restricting teachers to their own schedules and restricting administrative panels to IT administrators.	Protects employee privacy and prevents unauthorized schedule modifications.

NFR-COM-01	Compliance	The system shall retain immutable audit log records for at least 24 calendar months with SHA-256 verification hashes.	Satisfies regulatory labor record retention rules (Ministry of Manpower, 2026).
NFR-COM-02	Compliance	The system shall evaluate weekly working hours against the 44.0-hour statutory limit (Employment Act Part IV Section 38) and log overtime authorizations.	Mitigates institutional legal liability while supporting emergency overtime coverage.

2.3 Requirements Traceability Matrix (RTM)

Table 2.4: Requirements Traceability Matrix (RTM)

FR ID	Priority	Stakeholder Source	Mapped UC	Mapped SD	Mapped Class(es)
FR-01	M	Marcus Tan (MIN-01)	UC-01	SSD-01	ManagerDashboardUI, WorkloadController, Teacher, Schedule
FR-02	M	Marcus Tan (MIN-01)	UC-01	SSD-01	ManagerDashboardUI, WorkloadController, Teacher
FR-03	M	Marcus Tan (MIN-01)	UC-02	SSD-02	AllocationUI, AllocationController, ConflictValidator, Lesson
FR-04	M	Marcus Tan (MIN-01)	UC-02	SSD-02	AllocationUI, AllocationController, ConflictValidator, Teacher, Studio
FR-05	S	Marcus Tan (MIN-01)	UC-02	SSD-02	AllocationUI, WorkloadController, Teacher
FR-06	M	Elena Lim (MIN-02)	UC-07	SSD-07	TeacherUI, AvailabilityController, Availability, AuditLog
FR-07	M	Elena Lim (MIN-02)	UC-08	SSD-08	TeacherUI, PreferenceController, Preference, Teacher
FR-08	M	Elena Lim (MIN-02)	UC-09	SSD-09	TeacherUI, ScheduleController, Schedule, Teacher
FR-09	M	Elena Lim, Marcus Tan	UC-03, UC-10	SSD-03, SSD-10	TeacherUI, ManagerDashboardUI, RejectionController, AllocationController, Lesson, AuditLog
FR-10	S	Elena Lim (MIN-02)	UC-04	SSD-04	ManagerDashboardUI, AvailabilityController, Availability, Schedule
FR-11	C	Elena Lim (MIN-02)	UC-11	SSD-11	TeacherUI, ManagerDashboardUI, SwapController, ConflictValidator, Lesson
FR-12	S	Alex Wong (MIN-03)	UC-15	SSD-15	AdminUI, StudioController, Studio, MaintenanceWindow, Lesson
FR-13	S	Marcus Tan (MIN-01)	UC-06	SSD-06	AllocationUI, AllocationController, ConflictValidator, Lesson, Teacher

FR-14	M	Alex Wong (MIN-03)	UC-12, UC-13	SSD-12, SSD-13	AdminUI, AccountController, User, Teacher, Manager, ITAdministrator
FR-15	M	Alex Wong (MIN-03)	UC-14	SSD-14	AdminUI, StudioController, Studio, ConflictValidator
FR-16	M	Marcus Tan (MIN-01)	UC-05	SSD-05	AllocationUI, AllocationController, ConflictValidator, Schedule, AuditLog
FR-17	S	David Koh (MIN-05)	UC-16	SSD-16	ManagerDashboardUI, ReportController, Schedule, Studio
FR-18	M	David Koh (MIN-05)	UC-17	SSD-17	ManagerDashboardUI, ComplianceController, Teacher, Schedule
FR-19	M	David Koh (MIN-05)	UC-17	SSD-17	ManagerDashboardUI, ComplianceController, AuditLog

3. Use Cases & Activity Diagrams

3.1 Actor Identification & System Scope

The system boundary encompasses all workload planning, clash validation, availability tracking, reporting, and administrative governance. The system interfaces with three primary human actors:

- **Manager:** Allocates weekly lessons, reviews workload metrics, resolves late availability and rejection exceptions, handles emergency absence coverage, and publishes confirmed timetables.
- **Teacher:** Submits recurring monthly availability, updates schedule preferences, inspects personal confirmed timetables, initiates bilateral lesson swaps, and submits job rejection tickets.
- **IT Administrator:** Provisions user accounts, configures physical studio parameters, and schedules maintenance blackout windows.

UML Stereotype Relationships:

- `<<include>>` **(Mandatory Sub-behaviors):** The factored-out use case `Validate Allocation Constraints` is unconditionally executed whenever the base use case allocates or re-allocates a lesson, ensuring that every placement—whether a new assignment, a substitute re-assignment, or a peer swap—is validated against the same multi-constraint clash and fatigue rules. In accordance with UML standards, the dependency arrow points strictly from the base use case to the included use case (`Base ..> Included`).
    - `UC-02 (Allocate Weekly Teaching Job) <<include>> Validate Allocation Constraints`
    - `UC-03 (Review & Re-allocate Rejected Lesson) <<include>> Validate Allocation Constraints`
    - `UC-06 (Handle Emergency Absence Substitute) <<include>> Validate Allocation Constraints`
    - `UC-11 (Request Peer Lesson Swap) <<include>> Validate Allocation Constraints`
- `<<extend>>` **(Conditional Extension Points):** Optional behaviour is modelled as extensions that trigger only when a specific condition holds. In accordance with UML standards, the dependency arrow points from the extending use case back to the base use case (`Extending ..> Base`).
    - `Authorize Overtime <<extend>> UC-02 (Allocate Weekly Teaching Job)` — triggers when weekly hours enter the advisory tier (40.5h–44.0h), prompting the manager for a soft-warning override.
    - `Authorize Overtime <<extend>> UC-06 (Handle Emergency Absence Substitute)` — triggers when assigning a qualified substitute

pushes their weekly workload into the 40.5h–44.0h advisory range.

Non-Actor Boundary Decisions: In accordance with standard UML semantics, `School Management` is an external executive stakeholder receiving exported PDF reports out-of-band and is not a direct system actor. `Student` information is encapsulated as attributes within the `Lesson` entity; students do not directly log in or interact with the M1 system. Internal databases and notification services reside within the system boundary.

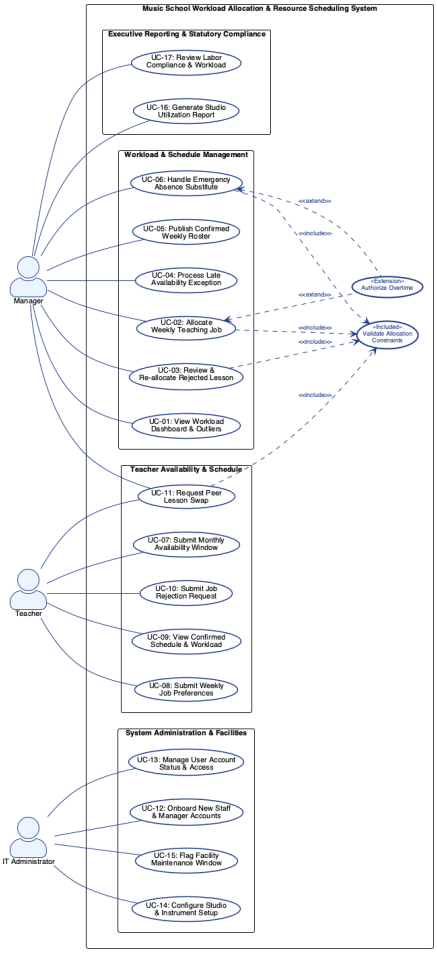


Figure 3.1: Master Use Case Diagram for Music School Scheduling System

3.2 Master Use Case Catalog Overview

Table 3.1: Master Use Case Catalog

UC ID	Use Case Title	Primary Actor	Priority	Operational Scope & Brief Summary
UC-01	View Workload Dashboard & Outliers	Manager	High	Real-time monitoring of teacher hours; highlights top 3 lowest-workload staff and >40h alerts.
UC-02	Allocate Weekly Teaching Job	Manager	High	Interactive allocation matrix comparing up to 3 teachers; sub-second clash validation.
UC-03	Review & Re-allocate Rejected Lesson	Manager	High	Exception queue management to reassign rejected lessons or release rooms.
UC-04	Process Late Availability Exception	Manager	Medium	Adjudication of post-19th availability revisions against preliminary schedule.
UC-05	Publish Confirmed Weekly Roster	Manager	High	Formal verification of unassigned slots, locking schedule status to Published, staff notifications.
UC-06	Handle Emergency Absence Substitute	Manager	Medium	Batch reallocation of scheduled lessons to substitutes due to acute instructor medical leave.
UC-07	Submit Monthly Availability Window	Teacher	High	Configuration of 30-min available slots up to 1 month ahead; editable until 19th 23:59 cutoff.
UC-08	Submit Weekly Job Preferences	Teacher	Medium	Capture of teaching day preferences, room acoustic preferences, and back-to-back clustering.
UC-09	View Confirmed Schedule & Workload	Teacher	High	Mobile-responsive timetable view displaying confirmed slots, running hours, and masked student PII.
UC-10	Submit Job Rejection Request	Teacher	High	Rejection initiation with mandatory consultation modal, reason codes, and queue routing.

UC-11	Request Peer Lesson Swap	Teacher	Low	Bilateral exchange workflow with collision checking, colleague acceptance, and manager approval.
UC-12	Onboard New Staff & Manager Accounts	IT Administrator	High	Account creation, credential assignment, and instrument qualification setup.
UC-13	Manage User Account Status & Access	IT Administrator	High	Logical deactivation, role modification, session revocation, and automated lesson relocation.
UC-14	Configure Studio & Instrument Setup	IT Administrator	Medium	Studio metadata configuration; acoustic baffling verification and Studio 5 drum exclusivity.
UC-15	Flag Facility Maintenance Window	IT Administrator	Medium	Scheduling blackout windows for piano tuning/servicing with automatic lesson relocation.
UC-16	Generate Studio Utilization Report	Manager	Medium	Aggregation of booked vs available studio hours across 86 weekly operating hours.
UC-17	Review Labor Compliance & Audit Trail	Manager	High	Audit of statutory 44h weekly caps, rest period verification, and immutable audit trail inspection.

3.3 Formal Use Case Specifications

3.3.1 Manager Operational Use Cases

UC-01: View Workload Dashboard & Outliers

- **Primary Actor:** Manager
- **Trigger:** Manager accesses system home landing page.
- **Priority:** High | **Secondary Actors:** None
- **Brief Description:** Manager monitors teacher workload distribution, identifying under-allocated teachers and teachers approaching or exceeding weekly thresholds.
- **Preconditions:** Manager is authenticated with active credentials.
- **Postconditions:**
  - *Main Success:* Real-time dashboard renders progress bars, top 3 lowest-workload teachers, and color-coded workload alert tiers.
  - *Alternative Failure:* System displays cached metrics with a synchronization warning if live data is temporarily unreachable.
- **Main Success Scenario:**
  1. Manager navigates to the Workload Dashboard view and selects the target calendar week.
  2. System queries current weekly schedule allocations across all active instructors and calculates running weekly and monthly hours (`runningWeeklyHours`, `runningMonthlyHours`) for each teacher.
  3. System renders the workload dashboard featuring:
     - Sorted list of the top 3 teachers with the lowest monthly allocated hours.
     - 3-tier visual badges: Green (<35.0h), Amber (35.0h–40.0h), and Red (>40.0h).
  4. Manager reviews the workload metrics and outlier rankings to plan upcoming allocations.
- **Alternative Scenarios:**
  - *3a. Teacher approaches 44.0h statutory ceiling:* System highlights teacher row with a Red statutory capacity badge (44.0h reached) and displays an advisory notice indicating that the teacher is at the statutory maximum and ineligible for additional lesson allocations.

UC-02: Allocate Weekly Teaching Job

- **Primary Actor:** Manager
- **Trigger:** Manager selects an unassigned or confirmed lesson slot to allocate.
- **Priority:** High | **Secondary Actors:** None
- **Brief Description:** Manager assigns a 30-minute lesson to an instructor and studio for a specific week, assisted by side-by-side comparison and instant constraint validation.
- **Preconditions:**
  - Target calendar week exists in `Draft` state; lesson slot is selected.

    - *Static & dynamic candidate invariants (enforced upfront by candidate and studio filtering, never offered as invalid options):* only teachers holding active certification for the lesson instrument whose projected weekly hours <= 44.0h are eligible candidates; Drums lessons can only be placed in Studio 5, as it is the sole studio with an acoustic drum kit.
- **Postconditions:**
    - *Main Success:* Lesson is bound to teacher and studio with status `Confirmed`, workload tallies increment by 0.5h, and transaction is logged.
    - *Alternative Failure:* Assignment is blocked; system displays specific constraint violation feedback.
- **Main Success Scenario:**
    1. Manager selects target week and student lesson slot.
    2. System retrieves lesson parameters (Instrument, Student Name, Timeslot, Duration = 30 min).
    3. System filters the candidate pool to teachers who are certified for the lesson instrument, marked available for the timeslot in their submitted availability, within the 4.0-hour continuous teaching limit (mandatory 1.0-hour rest break) if assigned, and whose projected weekly workload (current workload + 0.5h) <= 44.0h. System renders a candidate comparison panel displaying up to 3 of these teachers side-by-side with running weekly hours, monthly totals, preferences, and physical studio location for that date.
    4. Manager selects a candidate teacher. System restricts the studio picker to studios that are compatible with the lesson instrument (Studio 5 only for Drums), unoccupied, and not under a maintenance blackout window for the timeslot; all other studios are disabled. Manager selects an eligible studio.
    5. System re-validates the selected teacher, studio, and timeslot against the current schedule before saving.
    6. System evaluates projected teacher weekly hours (current workload + 0.5h):
        - If total hours <= 40.0h, assignment proceeds nominally.
        - If total hours > 40.0h and <= 44.0h (Advisory Tier), system prompts manager for soft-warning override and logs justification.
    7. System assigns lesson to teacher and studio, setting status to `Confirmed`.
    8. System increments teacher weekly and monthly workload tallies by 0.5h.
    9. System persists assignment and records immutable audit log entry.
    10. System renders updated schedule grid reflecting new booking.
- **Alternative Scenarios:**
    - *5a. Concurrent allocation conflict:* Another manager has booked the selected teacher or studio for the same timeslot after the options were displayed. System blocks placement displaying: "Allocation Conflict: Selection no longer available." and returns to step 3 with refreshed candidates and studios.

UC-03: Review & Re-allocate Rejected Lesson

- **Primary Actor:** Manager
- **Trigger:** System receives a teacher rejection ticket and flags the exception queue.
- **Priority:** High | **Secondary Actors:** Student, Substitute Teacher
- **Brief Description:** Manager reviews an instructor-rejected lesson, contacts the teacher if needed, and either rebinds the lesson to a qualified substitute or releases

the room and queues the student for priority rescheduling.

- **Preconditions:** Target lesson is in `Pending Re-allocation` state following teacher rejection.
- **Postconditions:**
  - *Main Success:* Lesson reassigned to qualified substitute teacher with status `Confirmed`, or released with status `Unassigned` and student priority queued.
  - *Alternative Failure:* Lesson remains in `Pending Re-allocation`; alert persists on dashboard.
- **Main Success Scenario:**
  1. Manager accesses the Exception Queue on the dashboard.
  2. System displays pending rejections with teacher ID, student, instrument, slot, reason code, and remarks. For each rejected lesson, the system auto-computes eligible substitute candidates via multi-tier pre-filtering:
     - *Static Pre-condition:* Only instructors holding active certification for the lesson instrument are considered.
     - *Dynamic Pre-condition:* Only instructors available during the 30-minute slot, whose continuous teaching block does not exceed 4.0h, and whose projected weekly workload <= 44.0h are returned.
     - Each rejection ticket is presented alongside its pre-computed candidate list.
  3. Manager reviews rejection ticket and pre-computed candidates, and contacts teacher if needed to explore resolution.
  4. Manager elects to reallocate lesson and selects a substitute from the pre-computed candidate list.
  5. If substitute's projected hours fall in the 40.5h–44.0h advisory range, manager confirms soft-warning override. System rebinds lesson to substitute, decrements original teacher hours by 0.5h, increments substitute hours by 0.5h, and transitions status to `Confirmed`.
  6. System dispatches confirmation notices to student and substitute teacher.
  7. System appends resolution outcome to immutable audit trail.
- **Alternative Scenarios:**
  - *3a. Consultation results in agreement to retain lesson:* Manager reinstates original teacher; system restores lesson status to `Confirmed` and logs resolution.
  - *2a. Zero qualified substitutes available for a rejection:* System displays "No eligible substitutes" on the rejection ticket. Manager releases studio to vacant pool. System transitions lesson status to `Unassigned`, places student on priority rescheduling list, dispatches cancellation notice to student, and logs audit record.

UC-04: Process Late Availability Exception

- **Primary Actor:** Manager
- **Trigger:** Teacher submits availability changes after the 19th 23:59 cutoff deadline.
- **Priority:** Medium | **Secondary Actors:** Teacher
- **Brief Description:** Manager inspects post-19th availability revisions against current preliminary schedules, deciding whether to approve or reject based on roster

disruption.

- **Preconditions:** Target submission has status `Late - Pending Manager Review`.
- **Postconditions:**
  - *Main Success:* Availability updated (`Approved Late`) or rejected (`Rejected`); teacher notified; audit log recorded.
- **Main Success Scenario:**
  1. Manager opens Late Availability Review queue.
  2. System displays diff comparing requested changes against preliminary roster.
  3. Manager evaluates operational disruption.
  4. Manager approves late availability with logged justification.
  5. System updates master availability schedule and transitions status to `Approved Late`.
  6. System dispatches approval notification to teacher and appends audit log.
- **Alternative Scenarios:**
  - *4a. Disruption unmanageable:* Manager rejects changes with explanatory notes. System maintains existing locked availability, sets status to `Rejected`, notifies teacher, and logs adjudication outcome.

UC-05: Publish Confirmed Weekly Roster

- **Primary Actor:** Manager
- **Trigger:** Manager completes weekly scheduling and initiates publishing.
- **Priority:** High | **Secondary Actors:** Teacher
- **Brief Description:** Manager executes final verification on a draft weekly timetable, locks allocations against informal edits, transitions status to `Published`, and releases confirmed rosters to teachers.
- **Preconditions:** Target week schedule is in `Draft` state.
- **Postconditions:**
  - *Main Success:* Schedule status transitions to `Published`; teacher personal timetables unlocked; audit record created.
- **Main Success Scenario:**
  1. Manager opens weekly schedule grid and initiates publishing review.
  2. System runs comprehensive verification checking for double-bookings, fatigue breaches, and unassigned lessons.
  3. System reports zero unassigned lessons and zero rule violations.
  4. Manager confirms publish command.
  5. System transitions schedule status to `Published` and locks direct manager allocation edits.
  6. System appends publication audit event and triggers notifications to all assigned instructors.
- **Alternative Scenarios:**
  - *3a. Unassigned lessons remain:* System renders warning banner detailing unassigned lesson count. Manager may either confirm publish with unassigned slots remaining (logged to audit) or abort publish to complete allocations.

UC-06: Handle Emergency Absence Substitute

- **Primary Actor:** Manager
- **Trigger:** Teacher reports acute medical or personal emergency for a scheduled shift.
- **Priority:** Medium | **Secondary Actors:** Teacher, Student
- **Brief Description:** Manager batch-reallocates scheduled lessons to qualified substitute teachers for an instructor on sudden leave, preserving lesson continuity.
- **Preconditions:** Target teacher has confirmed lessons on the specified date.
- **Postconditions:**
  - *Main Success:* Affected lessons rebound to substitutes or cancelled with student credit issued; audit trail recorded.
- **Main Success Scenario:**
  1. Manager inputs absent teacher ID, leave date, and reason code.
  2. System queries all confirmed lessons for that teacher on the leave date and auto-computes eligible substitute candidates for each lesson via `ConflictValidator` multi-tier pre-filtering:
     - *Static Pre-condition:* Only instructors holding active certification for the lesson instrument are considered.
     - *Dynamic Pre-condition:* Only instructors available during that 30-minute slot, whose continuous block does not exceed 4.0h, and whose projected weekly workload <= 44.0h are returned.
     - Each affected lesson is presented alongside its pre-computed candidate list.
  3. Manager reviews affected lessons and selects a qualified substitute from the pre-computed candidate list for each lesson.
  4. If substitute's workload enters the 40.5h–44.0h advisory range, manager confirms soft-warning override. System transfers lesson binding to substitute, updates respective workload hours (+/- 0.5h), transitions status to `Confirmed`, and logs audit event.
  5. Process repeats for all affected lessons; system notifies affected students and instructors.
- **Alternative Scenarios:**
  - *2a. No substitute available for a lesson:* System displays "No eligible substitutes" for that lesson. Manager selects "Cancel with Credit". System marks lesson `Cancelled`, issues student makeup credit, releases studio, and logs event.

3.3.2 Teacher Operational Use Cases

UC-07: Submit Monthly Availability Window

- **Primary Actor:** Teacher
- **Trigger:** Teacher accesses monthly availability grid before or on cutoff deadline.
- **Priority:** High | **Secondary Actors:** None
- **Brief Description:** Teacher configures recurring 30-minute available/unavailable timeslots up to 1 month ahead. The grid remains fully editable until 23:59 on the 19th of each month, after which changes route to manager review.
- **Preconditions:** Teacher is authenticated with active status.
- **Postconditions:**

    - *Main Success:* Availability persisted, timestamped, auto-approved (if <= 19th 23:59), and editable until cutoff.
    - *Alternative Failure:* Submissions after 19th 23:59 flagged as late and routed to manager queue.
- **Main Success Scenario:**
    1. Teacher opens Monthly Availability Grid.
    2. Teacher selects upcoming target month (up to 1 month ahead).
    3. Teacher toggles 30-minute slots indicating available/unavailable intervals across operating hours.
    4. Teacher reviews total available hours summary.
    5. Teacher clicks "Submit Availability" before 23:59 SGT on the 19th of the current month.
    6. System captures timestamp (<= 19th 23:59 SGT), sets status to `Auto-Approved`, persists records, and informs teacher that availability is confirmed and remains editable until the 19th cutoff.
    7. System logs submission in compliance audit trail.
- **Alternative Scenarios:**
    - *5a. Submission timestamp > 19th 23:59 SGT:* System sets status to `Late - Pending Manager Review`, locks direct grid editing, routes diff to Manager Exception Queue (`UC-04`), and notifies teacher that manager review is required.

UC-08: Submit Weekly Job Preferences

- **Primary Actor:** Teacher
- **Trigger:** Teacher accesses weekly preference configuration.
- **Priority:** Medium | **Secondary Actors:** None
- **Brief Description:** Teacher specifies pedagogical and scheduling preferences for an upcoming week, including preferred teaching days, studio acoustic preferences, and back-to-back lesson clustering.
- **Preconditions:** Teacher is authenticated; monthly availability has been submitted.
- **Postconditions:**
    - *Main Success:* Preferences persisted and linked to teacher profile to guide manager allocation tie-breaking.
- **Main Success Scenario:**
    1. Teacher navigates to Weekly Preferences panel.
    2. Teacher specifies preferred teaching days.
    3. Teacher toggles "Back-to-Back Lesson Clustering" to minimize unpaid breaks.
    4. Teacher selects preferred studio.
    5. Teacher submits preferences.
    6. System verifies preferences align with submitted availability and persists preference entity.
    7. System displays confirmation and logs preference update.

UC-09: View Confirmed Schedule & Workload

- **Primary Actor:** Teacher
- **Trigger:** Teacher opens personal timetable view.
- **Priority:** High | **Secondary Actors:** None

- **Brief Description:** Teacher reviews confirmed weekly timetable and running monthly workload hours on a responsive mobile interface with masked student PII.
- **Preconditions:** Teacher is authenticated.
- **Postconditions:**
  - *Main Success:* Timetable rendered with lesson times, student first names, instrument, and running monthly hours.
- **Main Success Scenario:**
  1. Teacher accesses schedule view on mobile or desktop browser.
  2. System queries teacher's lessons for target week.
  3. If schedule is `Published`, system displays confirmed lessons with student first name, studio, instrument, and time.
  4. System computes and displays running weekly hours and monthly total vs. requested target.
- **Alternative Scenarios:**
  - *3a. Schedule status is Draft:* System displays advisory notice: "Schedule currently in Draft - allocations pending manager publication."

UC-10: Submit Job Rejection Request

- **Primary Actor:** Teacher
- **Trigger:** Teacher clicks "Request Job Rejection" on a confirmed lesson slot.
- **Priority:** High | **Secondary Actors:** Manager
- **Brief Description:** Teacher initiates rejection of an assigned lesson slot. System displays a mandatory warning modal instructing the teacher to discuss the rejection with the manager before proceeding.
- **Preconditions:** Lesson is currently in `Confirmed` state and assigned to the teacher.
- **Postconditions:**
  - *Main Success:* Lesson transitions to `Pending Re-allocation`, urgent ticket routed to Manager Exception Queue, audit log created.
  - *Alternative Failure:* Teacher cancels at warning modal; assignment remains unchanged.
- **Main Success Scenario:**
  1. Teacher selects confirmed lesson slot in personal schedule.
  2. Teacher clicks "Request Job Rejection".
  3. System renders mandatory warning modal: "You must discuss this rejection with your manager before proceeding. Do you wish to continue?"
  4. Teacher confirms warning and clicks "Proceed with Rejection".
  5. Teacher selects standardized reason code (`Scheduling Conflict`, `Pedagogical Mismatch`, or `Personal / Medical`) and enters explanatory remarks.
  6. Teacher submits rejection ticket.
  7. System transitions lesson status to `Pending Re-allocation`.
  8. System routes urgent exception ticket to Manager Exception Queue (`UC-03`) and logs transaction.
  9. System displays confirmation acknowledging ticket submission.
- **Alternative Scenarios:**
  - *4a. Teacher declines warning modal:* Teacher clicks "Cancel"; system closes modal and maintains confirmed lesson without modifications.

UC-11: Request Peer Lesson Swap

- **Primary Actor:** Teacher
- **Trigger:** Teacher identifies a conflicting lesson and proposes bilateral swap with colleague.
- **Priority:** Low | **Secondary Actors:** Colleague Teacher, Manager
- **Brief Description:** Teacher proposes a bilateral lesson swap with another qualified instructor, subject to clash validation, colleague acceptance, and managerial approval.
- **Preconditions:** Both teachers hold confirmed lessons; both are certified for the respective instruments.
- **Postconditions:**
    - *Main Success:* Swap validated, approved by colleague and manager; lesson teacher assignments exchanged.
- **Main Success Scenario:**
    1. Teacher selects lesson and clicks "Propose Peer Swap".
    2. System filters candidate colleagues via multi-tier pre-filtering:
        - *Static Pre-condition:* Colleague holds active qualification for the lesson instrument.
        - *Dynamic Pre-condition:* Colleague is marked available during the target exchange slot, the swap does not cause either instructor to breach the 4.0-hour continuous teaching limit (minimum 1.0-hour rest break preserved), and neither teacher's projected weekly hours breach the 44.0h statutory ceiling.
    3. Teacher selects colleague and target exchange slot, submitting proposal.
    4. System executes dynamic safety validation ensuring zero timetable clashes or room collisions for either instructor.
    5. System notifies colleague teacher of pending swap proposal.
    6. Colleague teacher reviews and confirms swap proposal.
    7. System routes swap request to Manager for operational sign-off.
    8. Manager approves swap.
    9. System exchanges teacher assignments on both lessons, updates workload totals, and appends audit record.
- **Alternative Scenarios:**
    - *6a. Colleague teacher declines swap proposal:* Colleague rejects proposal; system notifies initiating teacher that swap was declined and maintains original lesson assignments.
    - *8a. Manager declines swap sign-off:* Manager rejects operational sign-off with recorded explanation; system notifies both instructors and maintains original lesson assignments.

3.3.3 IT Administrator Governance Use Cases

UC-12: Onboard New Staff & Manager Accounts

- **Primary Actor:** IT Administrator
- **Trigger:** New faculty or management personnel joins academy.
- **Priority:** High | **Secondary Actors:** None
- **Brief Description:** Administrator provisions user accounts for Teachers, Managers, or fellow Administrators, establishing access privileges and instrument qualifications.
- **Preconditions:** Administrator is authenticated with administrative privilege level.
- **Postconditions:**
  - *Main Success:* User entity created, role assigned, credentials generated, audit log appended.
- **Main Success Scenario:**
  1. Administrator accesses User Management Console.
  2. Administrator enters staff particulars (Full Name, Staff ID, School Email, Contact).
  3. Administrator selects role (`Teacher`, `Manager`, or `IT Administrator`).
  4. If role is `Teacher`, administrator selects certified instruments (`Piano`, `Drums`, `Violin`, `Trumpet`).
  5. System verifies uniqueness of Staff ID and email.
  6. System creates account in active status and logs provisioning event.
- **Alternative Scenarios:**
  - *5a. Duplicate Staff ID or Email:* System halts creation and displays error: "Duplicate account detected."

UC-13: Manage User Account Status & Access

- **Primary Actor:** IT Administrator
- **Trigger:** Staff departure, leave of absence, or credential review.
- **Priority:** High | **Secondary Actors:** Manager
- **Brief Description:** Administrator modifies user role or executes logical account deactivation.
- **Preconditions:**
  - Administrator is authenticated; target user account exists.
  - *Static invariant (design constraint):* The interface offers only logical deactivation. No physical deletion control is provided, so immutable audit history is always retained.
- **Postconditions:**
  - *Main Success:* Account status set to inactive, sessions revoked, future lessons shifted to `Pending Re-allocation`.
- **Main Success Scenario:**
  1. Administrator searches for target user account.
  2. Administrator selects "Deactivate Account".
  3. System prompts confirmation warning detailing active future lesson allocations.
  4. Administrator confirms deactivation.
  5. System marks the account inactive and revokes active login sessions.

6. System unassigns all future lessons held by the user, moving them to `Pending Re-allocation` on the manager queue.
7. System logs administrative action in immutable audit log.

UC-14: Configure Studio & Instrument Setup

- **Primary Actor:** IT Administrator
- **Trigger:** Room reconfiguration or instrument acquisition.
- **Priority:** Medium | **Secondary Actors:** None
- **Brief Description:** Administrator configures room parameters for Studios 1–5, enforcing acoustic drum exclusivity in Studio 5.
- **Preconditions:**
  - Administrator is authenticated.
  - *Static invariant (hardware registry):* The acoustic drum kit is hard-configured to Studio 5's asset profile (acoustic baffling). Studios 1–4 are piano-only rooms in the registry.
- **Postconditions:**
  - *Main Success:* Studio parameters persisted and propagated to scheduling engine within 500ms.
- **Main Success Scenario:**
  1. Administrator opens Studio Configuration Panel.
  2. System displays registry for Studios 1–5 with installed instrument metadata and operating hours (Mon–Fri 09:00–21:00, Sat–Sun 08:00–21:00).
  3. Administrator configures editable studio attributes (operating hours, piano and general instrument metadata).
  4. System validates that the entered operating hours are valid and that the change does not conflict with confirmed lessons.
  5. System persists configuration, updates validation engine cache, and logs event.

UC-15: Flag Facility Maintenance Window

- **Primary Actor:** IT Administrator
- **Trigger:** Scheduled instrument servicing (eg piano tuning) or emergency facility repair.
- **Priority:** Medium | **Secondary Actors:** Manager
- **Brief Description:** Administrator schedules a maintenance blackout window for a studio, automatically suppressing the room from the allocation matrix and relocating conflicting lessons.
- **Preconditions:** Administrator is authenticated; target studio selected.
- **Postconditions:**
  - *Main Success:* Maintenance window persisted; studio suppressed from matrix; conflicting lessons shifted to `Pending Re-allocation`.
- **Main Success Scenario:**
  1. Administrator selects target studio and specifies blackout start/end times.
  2. Administrator enters servicing justification (eg "Semi-Annual Piano Tuning").
  3. System scans window for existing confirmed lessons.
  4. System flags blackout window, suppressing room from manager allocation grid.

5. If lessons exist during window, system shifts them to `Pending Re-allocation` and notifies Manager.
6. Administrator confirms blackout entry; system persists maintenance record and logs audit event.

- **Alternative Scenarios:**
  - *1a. Emergency repair (eg water leak):* Administrator clicks "Emergency Blackout"; system immediately locks studio, cancels active sessions in that room, and issues immediate alerts to on-duty managers.

3.3.4 Executive Reporting & Compliance Use Cases

UC-16: Generate Studio Utilization Report

- **Primary Actor:** Manager
- **Trigger:** End of monthly scheduling cycle or management request.
- **Priority:** Medium | **Secondary Actors:** School Management (External)
- **Brief Description:** Manager generates and exports studio occupancy analytics across all 86 weekly operating hours for executive governance.
- **Preconditions:** Manager is authenticated; schedule data persisted.
- **Postconditions:**
  - *Main Success:* System compiles utilization percentages, peak hour distributions, and exports report.
- **Main Success Scenario:**
  1. Manager opens Analytics & Reporting view.
  2. Manager selects target month.
  3. System aggregates lesson hours across Studios 1–5, computing booked hours against monthly capacity (344h per studio, 1,720h academy total).
  4. System renders utilization charts and peak window heatmaps.
  5. Manager exports report in PDF/CSV format for executive presentation.

UC-17: Review Labor Compliance & Audit Trail

- **Primary Actor:** Manager
- **Trigger:** Weekly pre-publish compliance and governance audit.
- **Priority:** High | **Secondary Actors:** None
- **Brief Description:** Manager audits faculty working hours against the statutory 44.0h standard under the Singapore Employment Act 1968, inspects daily rest compliance, and queries the immutable audit log (`AuditLog`) to verify overrides and administrative actions.
- **Preconditions:** Manager is authenticated; teacher workload tallies and audit records are persisted.
- **Postconditions:**
  - *Main Success:* Compliance report and filtered audit trail rendered, statutory overtime verified, sign-off logged to audit trail.
- **Main Success Scenario:**
  1. Manager navigates to the Labor Compliance & Audit Trail panel and selects the target calendar week.
  2. System evaluates teacher weekly hours against statutory standards (Employment Act Part IV Section 38) and verifies that zero instructors exceed the 44.0h statutory ceiling.
  3. Manager selects the Audit Trail filter (`getAuditTrail`) to inspect immutable `AuditLog` entries for lesson allocations, advisory overtime overrides (40.5h–44.0h), rejection resolutions, and administrative updates.
  4. System retrieves and displays matching `AuditLog` records showing timestamp, actor ID, action type, and justification details.
  5. Manager inspects daily workload distribution ensuring no teacher exceeds 12 daily hours and submits the weekly compliance verification snapshot; system

appends an immutable sign-off record to `AuditLog`.

- **Alternative Scenarios:**
  - *3a. Missing justification for advisory overtime (40.5h–44.0h):* If an instructor was allocated within the advisory overtime range without a complete managerial justification text, system prompts manager to review the operational context and append the justification record before allowing compliance sign-off.

3.4 Dynamic Process Modeling: Activity Diagrams

3.4.1 Flagship Activity Workflow: Weekly Job Allocation

Figure 3.2 illustrates the complete job allocation workflow executed by the Manager, detailing upfront candidate and studio filtering, save-time re-validation, workload threshold evaluation, lesson assignment, and immutable audit logging.

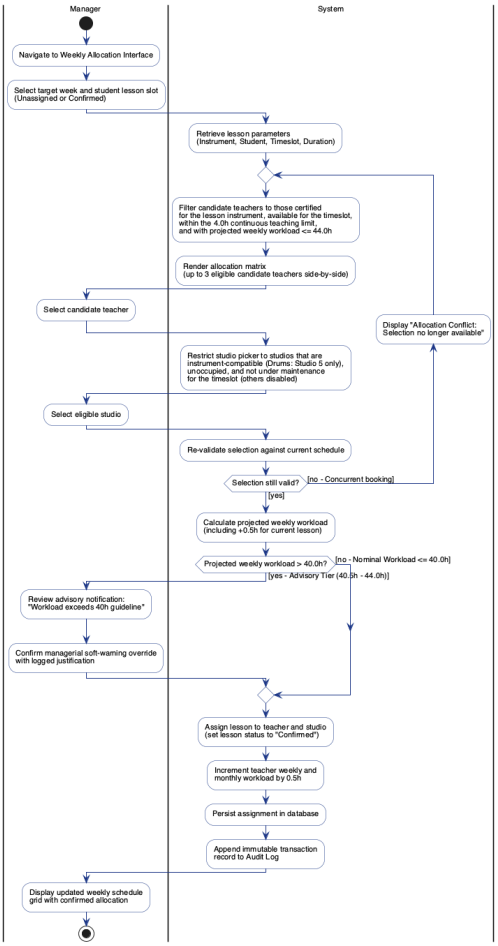


Figure 3.2: Activity Diagram for Weekly Job Allocation Workflow

3.4.2 2-Stage Job Rejection Workflow

Figure 3.3 illustrates the 2-stage rejection lifecycle, showing the mandatory teacher consultation modal, reason code capture, manager exception review, substitute reassignment, and audit recording.

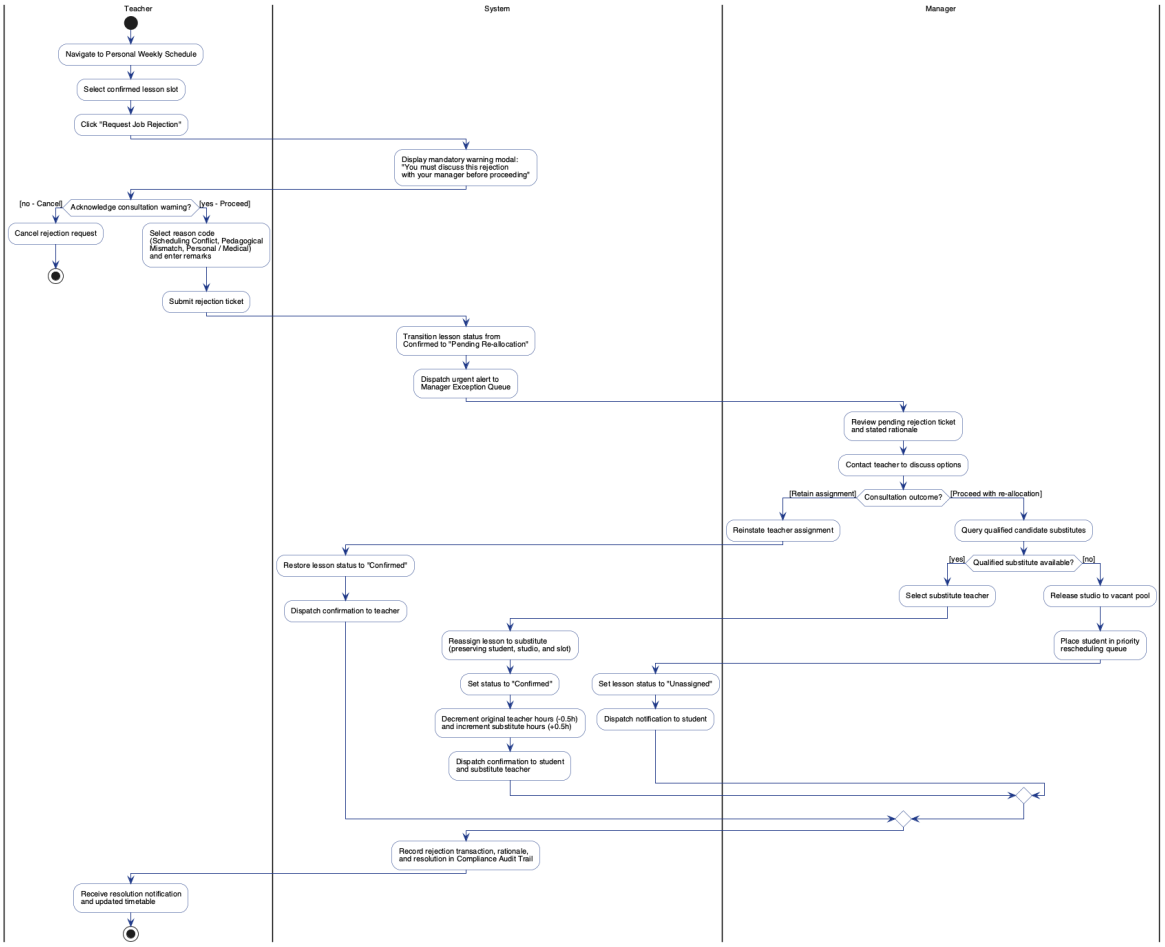


Figure 3.3: Activity Diagram for 2-Stage Job Rejection Workflow

3.4.3 Monthly Availability Cutoff & Exception Review Workflow

Figure 3.4 models the monthly availability workflow, showing on-time submissions (editable until 19th 23:59), post-cutoff manager exception routing, adjudication decisions, and convergence into the compliance audit trail.

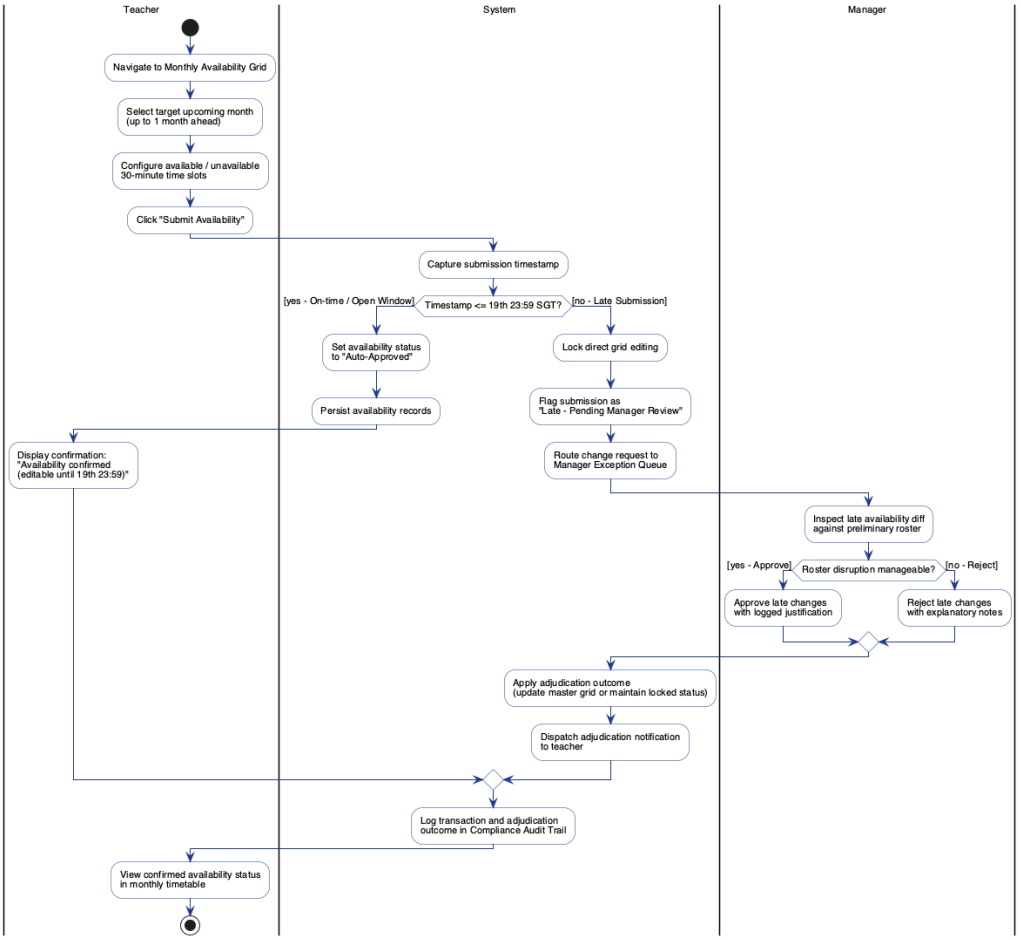


Figure 3.4: Activity Diagram for Availability Cutoff & Exception Review

4. Object-Oriented Analysis

4.1 4-Step Object-Oriented Derivation

4.1.1 Step 1: Solution Abstract (Domain Narrative)

The client music academy operates five physical studios (Studio 1 to Studio 5) across 86 weekly operating hours (Mon–Fri 09:00–21:00, Sat–Sun 08:00–21:00), providing individual 30-minute lessons across Piano, Drums, Violin, and Trumpet. The academy serves 251 enrolled students, staffed by nine teachers, two shift managers, and IT administrators. The web-based system replaces spreadsheet rosters with a visual decision-support platform. Human managers allocate weekly jobs assisted by sub-second clash validation. Availability must be submitted by 23:59 on the 19th of each month (planning begins on the 20th); late submissions route to manager review. Physical constraints dictate that all studios house pianos, but only Studio 5 houses an acoustic drum kit. Pedagogical safety requires that instructors teach at most 4.0 continuous hours before receiving a mandatory 1.0-hour break. Statutory rules monitor the 44-hour weekly threshold under Singapore's Employment Act 1968, tracking authorized overtime at 1.5x. All mutations, rejections, and managerial overrides are recorded in an append-only audit trail retained for 24 calendar months.

4.1.2 Step 2: Noun Filtering & Candidate Class Elimination Matrix

Candidate Domain Noun	Initial Domain Role	Modeling Evaluation & Decision	Final Stereotype & Architectural Rationale
Excel Spreadsheet	Legacy Medium	**Eliminated**	Replaced entirely by web application database persistence.
Classroom Lights / AC	Facility Hardware	**Eliminated**	Environmental facilities not automated or governed by scheduling logic.
Central Database	System Storage	**Eliminated**	Data store is an internal passive mechanism, omitted from domain model.
Public Holiday	Calendar Rule	**Eliminated**	Governed as business validation rule enforced by scheduling controllers.
Waitlist / Intake	Enrollment Record	**Eliminated**	Student admissions handled out-of-band; excluded from M1 workload scope.

Timeslot / Job	Temporal Unit	**Attribute**	Represented as `startDateTime` and `endDateTime` attributes in `Lesson`.
Student	Direct Customer	**Attribute**	Student data encapsulated as `studentId` and `studentName` in `Lesson`.
ManagerDashboardUI	Presentation View	**Concrete Class**	`<<Boundary>>` Renders workload progress bars, outliers, and compliance panels.
AllocationUI	Presentation View	**Concrete Class**	`<<Boundary>>` Facilitates side-by-side teacher comparison and slot binding.
TeacherUI	Presentation View	**Concrete Class**	`<<Boundary>>` Renders schedule, availability grid, preferences, rejection modal.
AdminUI	Presentation View	**Concrete Class**	`<<Boundary>>` Renders user management, studio configuration, and blackouts.
WorkloadController	Manpower Logic	**Concrete Class**	`<<Control>>` Computes hours, sorts lowest-workload staff, audits 40h warnings.
AllocationController	Scheduling Orchestrator	**Concrete Class**	`<<Control>>` Coordinates lesson binding, hour updates, persistence, and audit.
ConflictValidator	Rule Enforcement	**Concrete Class**	`<<Control>>` Pre-filters eligible candidates (instrument skills, availability, 4h block, <= 44.0h) and studios (Studio 5 drum rule, occupancy, maintenance); re-validates at save.
AvailabilityController	Cutoff Enforcement	**Concrete Class**	`<<Control>>` Validates availability submissions and enforces 19th 23:59 cutoff.
RejectionController	Rejection Workflow	**Concrete Class**	`<<Control>>` Processes job rejection requests and routes to manager queue.

SwapController	Peer Swap	**Concrete Class**	`<<Control>>` Orchestrates bilateral peer swap verification and approvals.
AccountController	User Lifecycle	**Concrete Class**	`<<Control>>` Provisions and deactivates accounts; enforces RBAC permissions.
StudioController	Facility Management	**Concrete Class**	`<<Control>>` Manages studio metadata, operating hours, and maintenance windows.
ReportController	Analytics	**Concrete Class**	`<<Control>>` Aggregates studio utilization metrics across 86 operating hours.
ComplianceController	Labor Compliance	**Concrete Class**	`<<Control>>` Audits statutory 44h weekly caps, 12h daily spread, and overtime logs.
ScheduleController	Schedule Retrieval	**Concrete Class**	`<<Control>>` Retrieves teacher schedules and published weekly rosters.
PreferenceController	Preference Validation	**Concrete Class**	`<<Control>>` Validates teacher preferences against submitted availability.
User (Abstract)	Core Identity	**Abstract Class**	`<<Entity>>` Abstract superclass encapsulating staff ID, name, email, and role.
Teacher	Human Resource	**Concrete Class**	`<<Entity>>` Specializes `User`; stores qualifications, availability, and running hours.
Manager	Human Resource	**Concrete Class**	`<<Entity>>` Specializes `User`; tracks shift assignment and managed operations.
ITAdministrator	Systems Governance	**Concrete Class**	`<<Entity>>` Specializes `User`; maintains privilege levels for administration.
Lesson	Pedagogical Unit	**Concrete Class**	`<<Entity>>` 30-min unit binding student, teacher, studio, and status lifecycle.

Studio	Physical Room	**Concrete Class**	`<<Entity>>` Studios 1–5, tracking piano/drum assets, operating hours, blackouts.
MaintenanceWindow	Facility Service	**Concrete Class**	`<<Entity>>` Specific servicing intervals during which a studio is suppressed.
Availability	Planning Input	**Concrete Class**	`<<Entity>>` 30-min availability blocks, submission timestamp, approval status.
Preference	Teacher Priority	**Concrete Class**	`<<Entity>>` Preferred teaching days, studio selection, and back-to-back clustering.
RejectionRequest	Operational Exception	**Concrete Class**	`<<Entity>>` Captures reason code, remarks, and resolution status for a rejection.
Schedule	Weekly Aggregate	**Concrete Class**	`<<Entity>>` Aggregates lessons for calendar week and manages Draft/Published states.
AuditLog	Regulatory Compliance	**Concrete Class**	`<<Entity>>` Immutable audit record satisfying MOM 24-month retention requirement.

4.1.3 Step 3: Class-Responsibility-Collaborator (CRC) ECB Matrix

ECB Tier	Class Name	Primary Operational Responsibilities	Collaborators
Boundary	**ManagerDashboardUI**	Render workload progress bars, capacity warnings, lowest 3 staff, utilization panels.	WorkloadController, ReportController, ComplianceController
Boundary	**AllocationUI**	Render side-by-side 3-teacher view, studio picker, display real-time validation errors.	AllocationController

Boundary	**TeacherUI**	Render schedule, availability grid, preferences, rejection modal, swap requests.	AvailabilityController, ScheduleController, RejectionController, PreferenceController, SwapController
Boundary	**AdminUI**	Render user management console, studio configuration panels, maintenance modals.	AccountController, StudioController
Control	**WorkloadController**	Calculate weekly/monthly hours, rank workload deficit, flag >40h warning staff.	Teacher, Schedule
Control	**AllocationController**	Orchestrate lesson binding, hour updates, persistence, and audit logging.	ConflictValidator, Lesson, Schedule, AuditLog
Control	**ConflictValidator**	Pre-filter eligible candidates (instrument skills, availability, 4h block, <= 44.0h) and studios (Studio 5 drum rule, occupancy, maintenance); re-validate at save.	Teacher, Studio, Schedule
Control	**AvailabilityController**	Validate availability submissions, enforce 19th 23:59 cutoff, route late submissions.	Availability, AuditLog
Control	**RejectionController**	Process rejection requests, transition lesson status, alert manager exception queue.	Lesson, RejectionRequest, AuditLog
Control	**SwapController**	Validate peer swap requests, verify clash rules, record manager sign-off.	ConflictValidator, Lesson, AuditLog

Control	**AccountController**	Provision and deactivate accounts, assign role permissions, relocate lessons.	User, Lesson, AuditLog
Control	**StudioController**	Configure studios, schedule maintenance windows, relocate affected lessons.	Studio, MaintenanceWindow, Lesson, AuditLog
Control	**ReportController**	Aggregate studio utilization metrics across 86 operating hours.	Schedule, Studio
Control	**ComplianceController**	Audit statutory 44h weekly caps, 12h daily limits, and record overtime justifications.	Teacher, Schedule, AuditLog
Control	**ScheduleController**	Retrieve teacher timetables and published weekly schedules.	Schedule, Teacher
Control	**PreferenceController**	Validate weekly job preferences against submitted availability windows.	Preference, Teacher
Entity	**User (Abstract)**	Superclass maintaining staffId, name, email, role, and active account status.	AuditLog
Entity	**Teacher**	Specializes User; stores instrument qualifications, availability, and workloads.	Availability, Preference, Lesson
Entity	**Manager**	Specializes User; tracks managed shift hours and shift schedules.	User, AuditLog

Entity	**ITAdministrator**	Specializes User; maintains administrative privilege credentials.	User, AuditLog
Entity	**Lesson**	30-min instructional unit binding student, teacher, studio, and lifecycle status.	Teacher, Studio, RejectionRequest
Entity	**Studio**	Physical room tracking piano/drum assets, operating hours, and servicing state.	MaintenanceWindow, Lesson
Entity	**MaintenanceWindow**	Defines scheduled servicing blackout intervals with start and end timestamps.	Studio
Entity	**Availability**	Stores recurring 30-min available slots and tracks approval status lifecycle.	Teacher
Entity	**Preference**	Records preferred teaching days, studio selection, and back-to-back clustering.	Teacher
Entity	**RejectionRequest**	Stores rejection reason code, explanatory remarks, and resolution outcome.	Lesson
Entity	**Schedule**	Aggregates lessons for calendar week and manages Draft/Published states.	Lesson, Studio
Entity	**AuditLog**	Append-only immutable log recording actor, action, timestamp, and details.	User, Schedule, Lesson

4.1.4 Step 4: Relationship Derivation

- **Generalization (Inheritance):** `User` (abstract) is the superclass for `Teacher`, `Manager`, and `ITAdministrator`, which inherit `staffId`, `fullName`, `email`, `role`, and `isActive`. `Teacher` specializes with `instrumentQualifications`

and workload methods. `Manager` specializes with `managedShiftHours`. `ITAdministrator` specializes with `privilegeLevel`.

- **Composition:** `Schedule` composes `Lesson` (`1` to `0..*`) because lessons cannot exist independently of a schedule. `Teacher` composes `Availability` (`1` to `0..*`) and `Preference` (`1` to `0..1`). `Studio` composes `MaintenanceWindow` (`1` to `0..*`). `Lesson` composes `RejectionRequest` (`1` to `0..1`).
- **Association:** `Lesson` is associated with `Teacher` (`0..*` to `0..1`, role `assignedTo`) and `Studio` (`0..*` to `0..1`, role `conductedIn`). `User` initiates `AuditLog` records (`1` to `0..*`), which target `Schedule` and record `Lesson` mutations.
- **Dependency:** Strict Entity-Control-Boundary layering is enforced: Boundary classes depend only on Control classes (invokes). Boundary `AllocationUI` invokes `AllocationController`, which delegates rule validation to `ConflictValidator` (validates via). `SwapController` delegates to `ConflictValidator`. Boundary `ManagerDashboardUI` invokes `WorkloadController`, `ReportController`, and `ComplianceController`. Human actors never call Control or Entity classes directly.

4.2 Domain Class Diagram

The class model is partitioned into two clear views: Figure 4.1 presents the structural domain entity model (inheritance, compositions, associations, and enumerations), while Figure 4.2 details the boundary-control architectural dependency map.

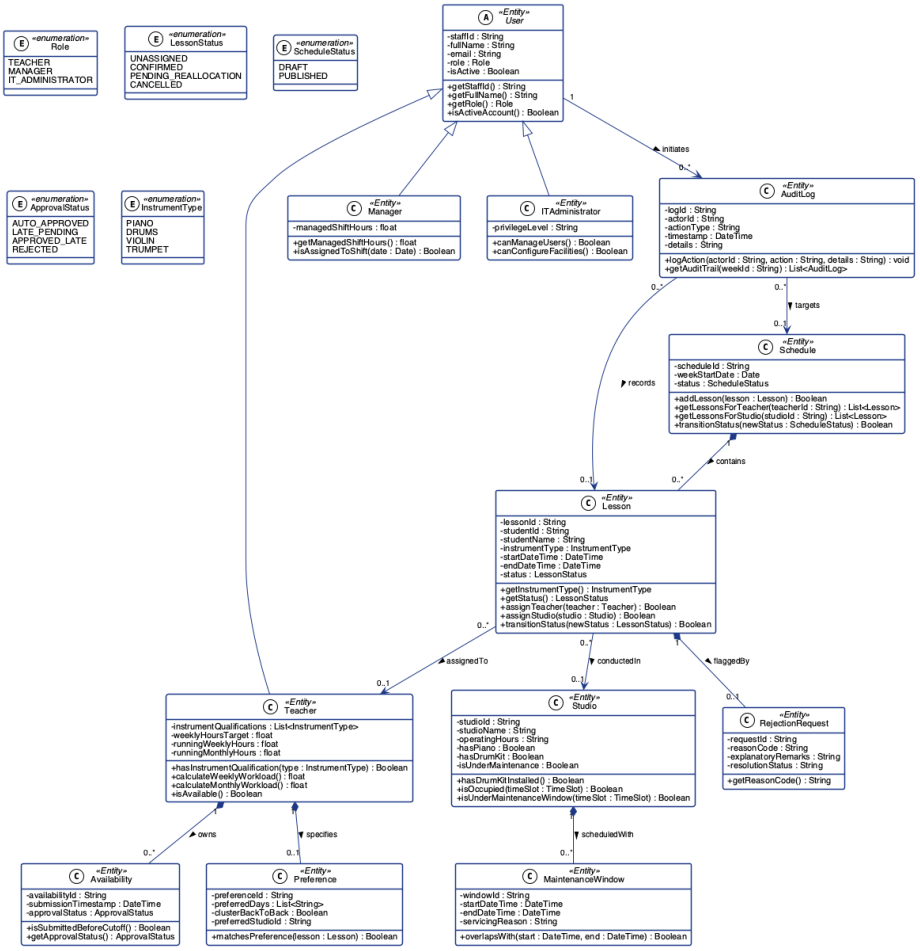


Figure 4.1: Domain Entity Class Diagram (Inheritance, Compositions, Associations & Enums)

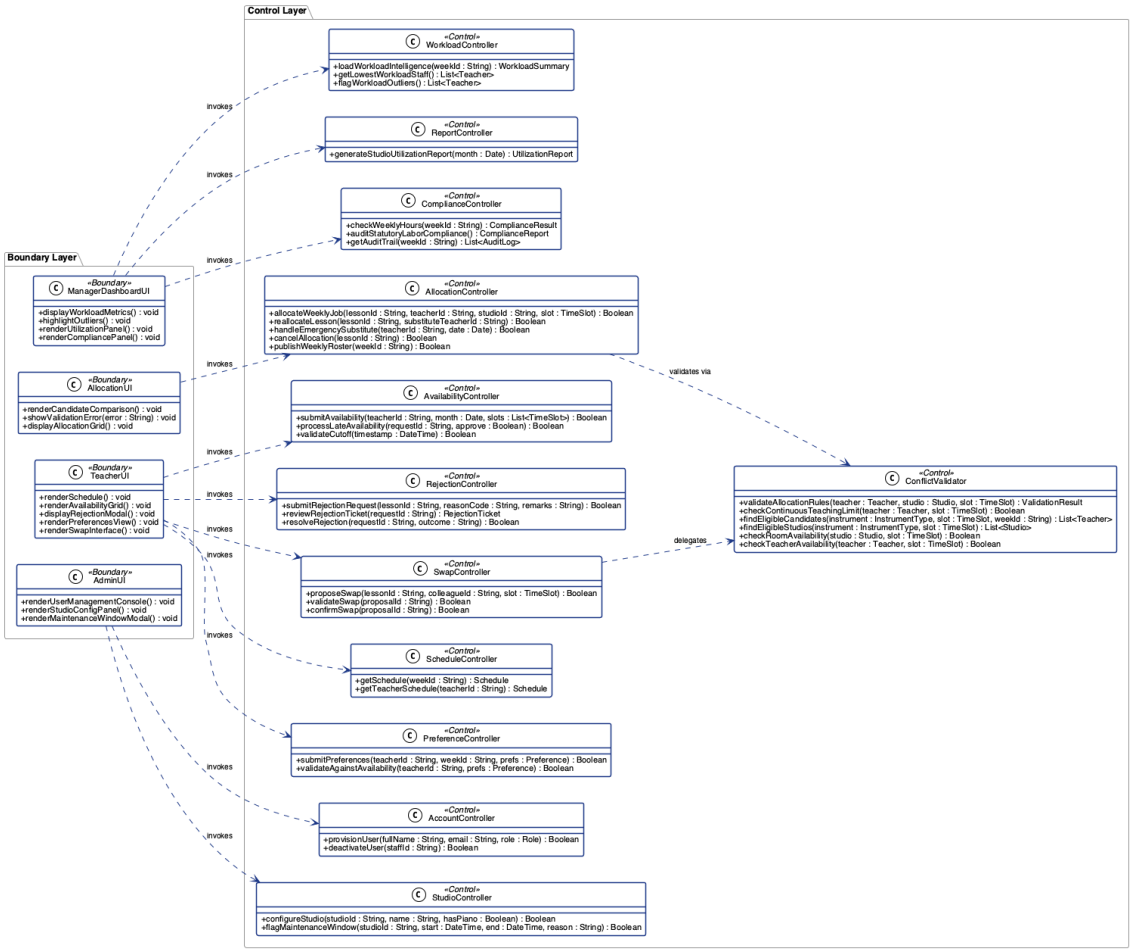


Figure 4.2: Boundary-Control Architectural Dependency Map (ECB Invocations & Delegations)

4.3 Sequence Diagrams

The 17 detailed sequence diagrams below model dynamic interaction flows across the Entity-Control-Boundary architecture, showing synchronous calls, activations, return messages, combined fragments, and immutable audit logging.

SSD-01: View Workload Dashboard & Outliers (UC-01)

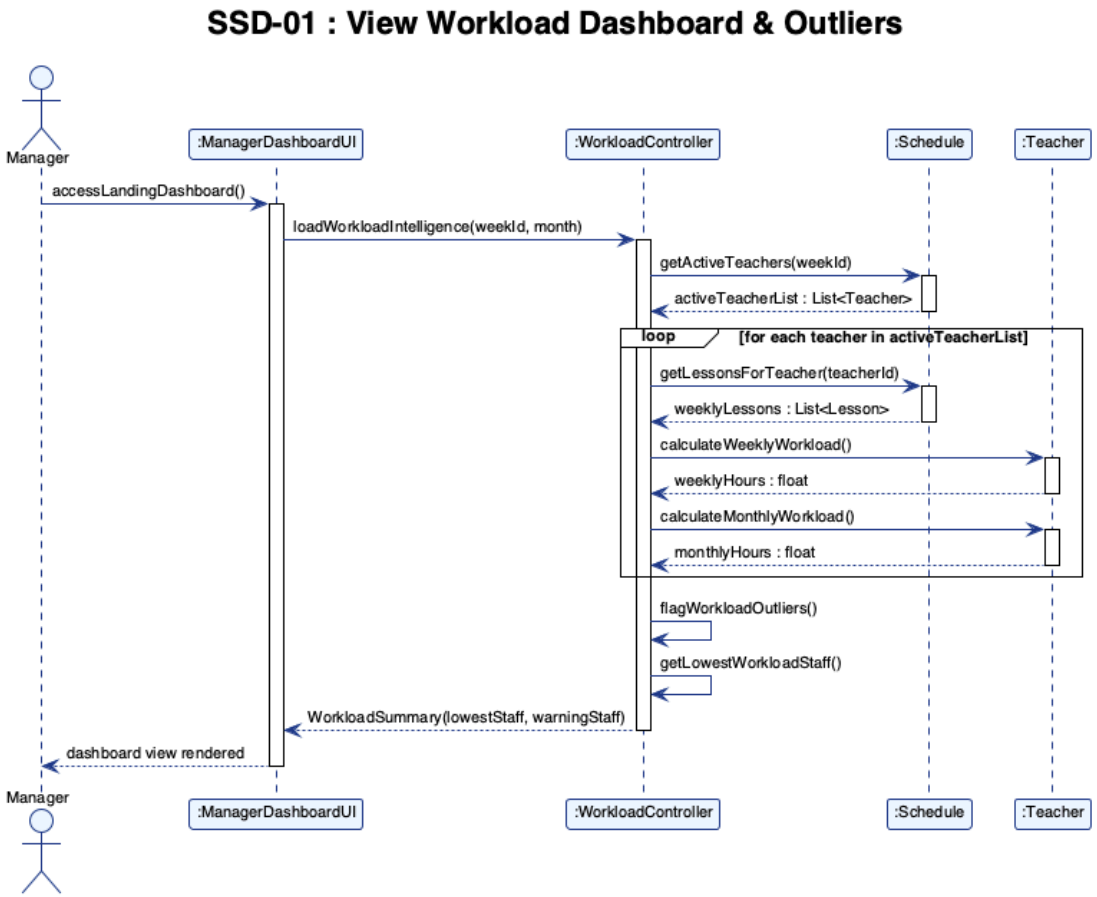


Figure 4.3: SSD-01 View Workload Dashboard & Outliers

SSD-02: Allocate Weekly Teaching Job (UC-02)

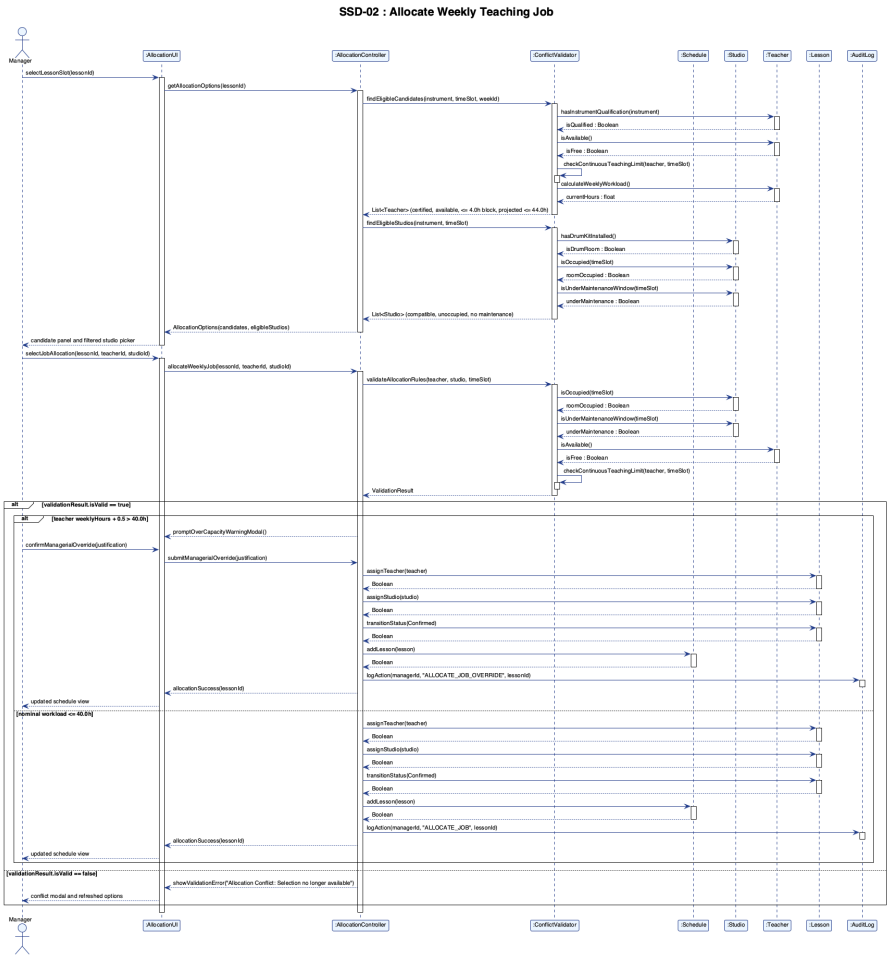


Figure 4.4: SSD-02 Allocate Weekly Teaching Job

SSD-03: Review & Re-allocate Rejected Lesson (UC-03)

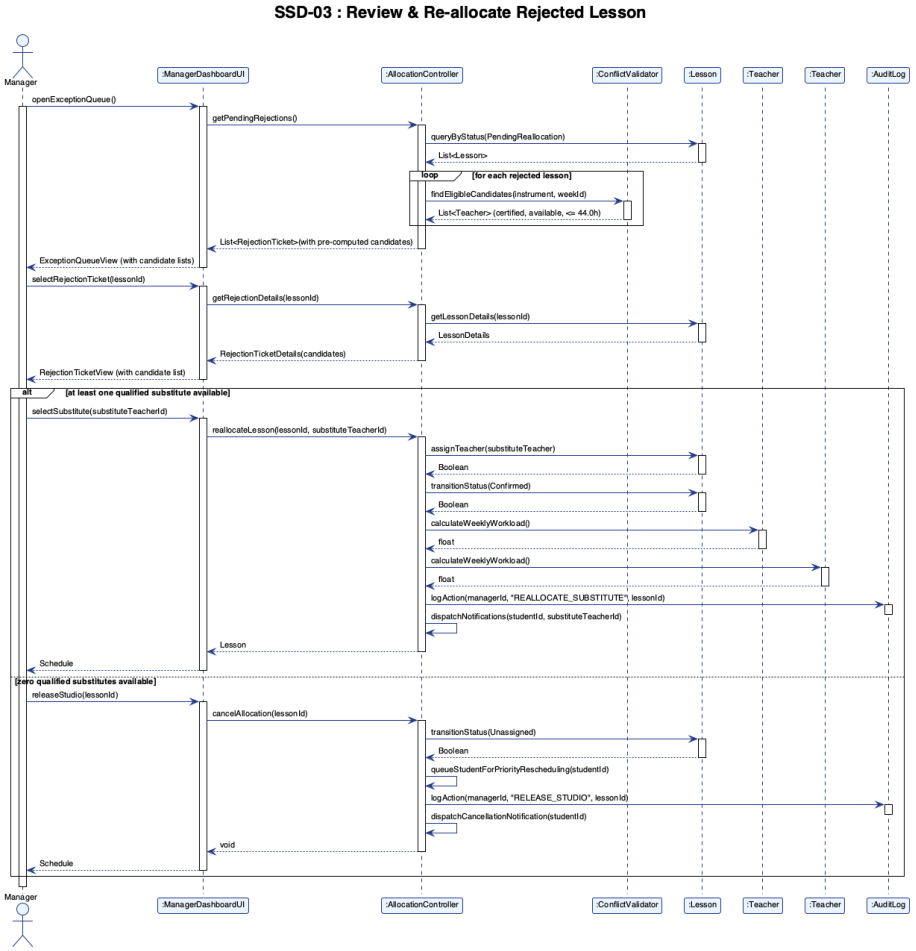


Figure 4.5: SSD-03 Review & Re-allocate Rejected Lesson

SSD-04: Process Late Availability Exception (UC-04)

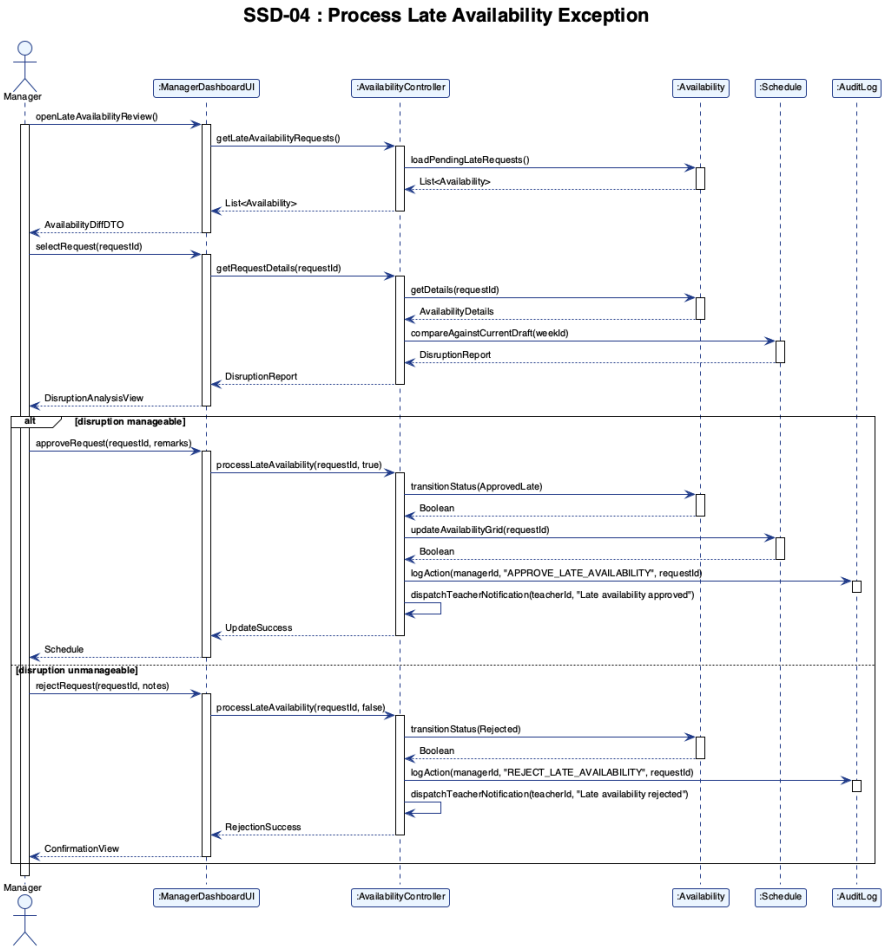


Figure 4.6: SSD-04 Process Late Availability Exception

SSD-05: Publish Confirmed Weekly Roster (UC-05)

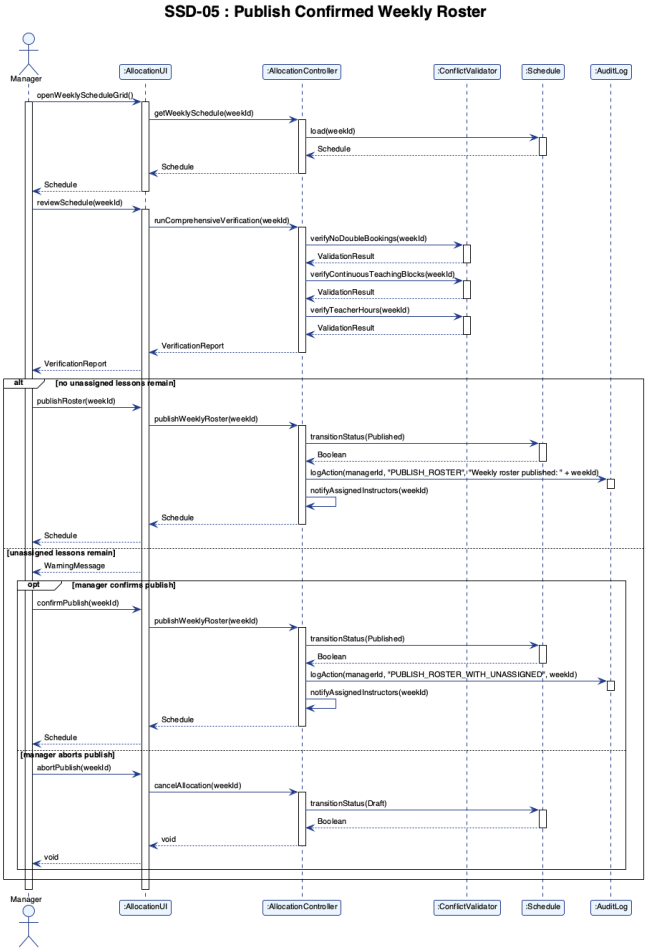


Figure 4.7: SSD-05 Publish Confirmed Weekly Roster

SSD-06: Handle Emergency Absence Substitute (UC-06)

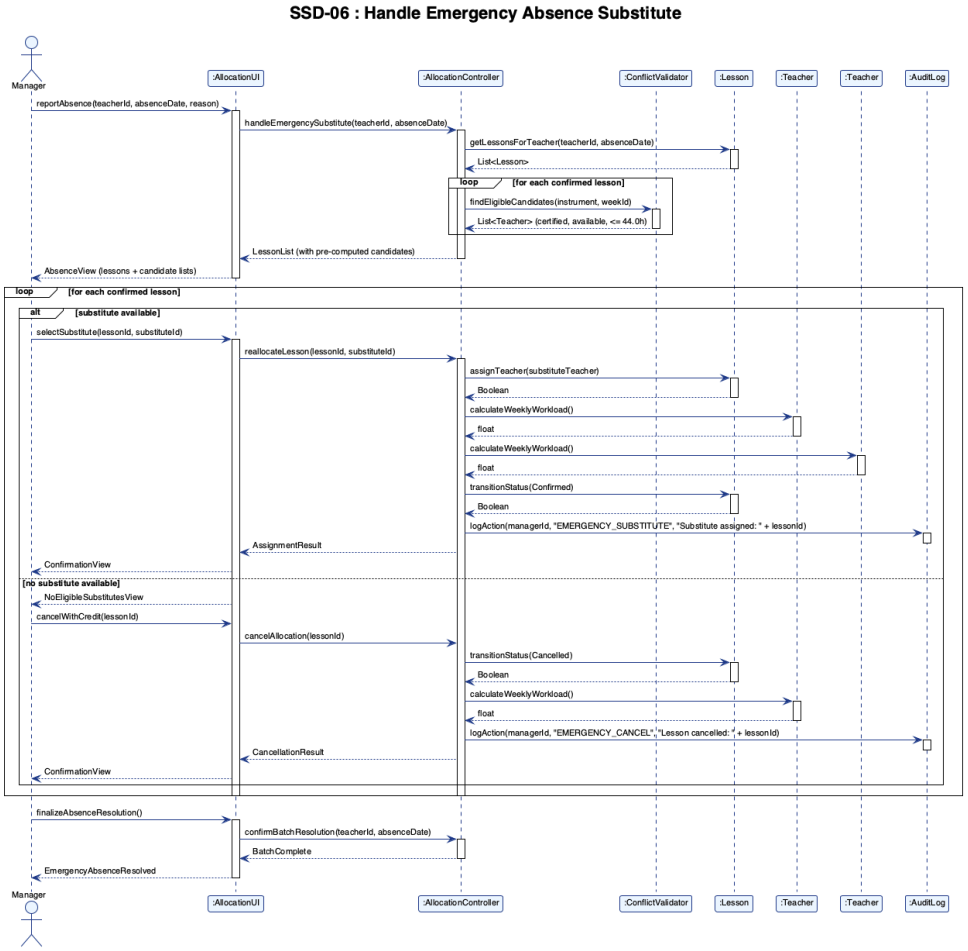


Figure 4.8: SSD-06 Handle Emergency Absence Substitute

SSD-07: Submit Monthly Availability Window (UC-07)

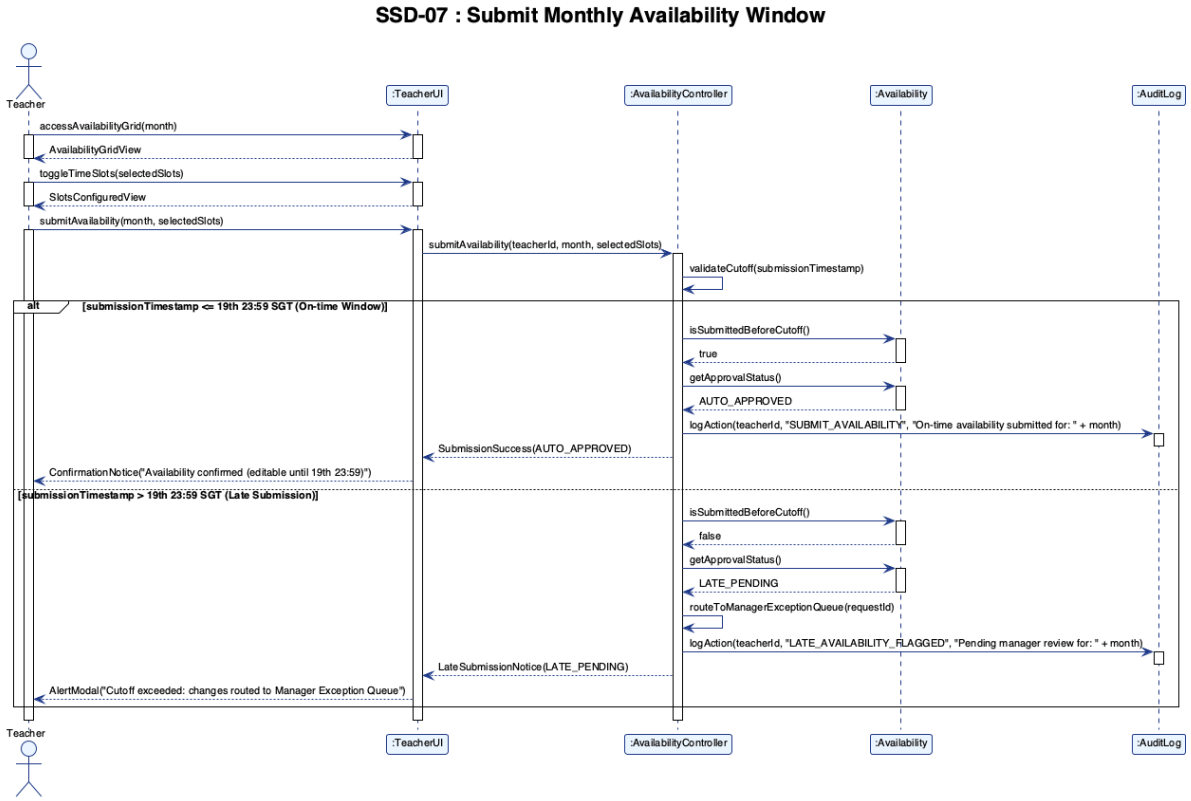


Figure 4.9: SSD-07 Submit Monthly Availability Window

SSD-08: Submit Weekly Job Preferences (UC-08)

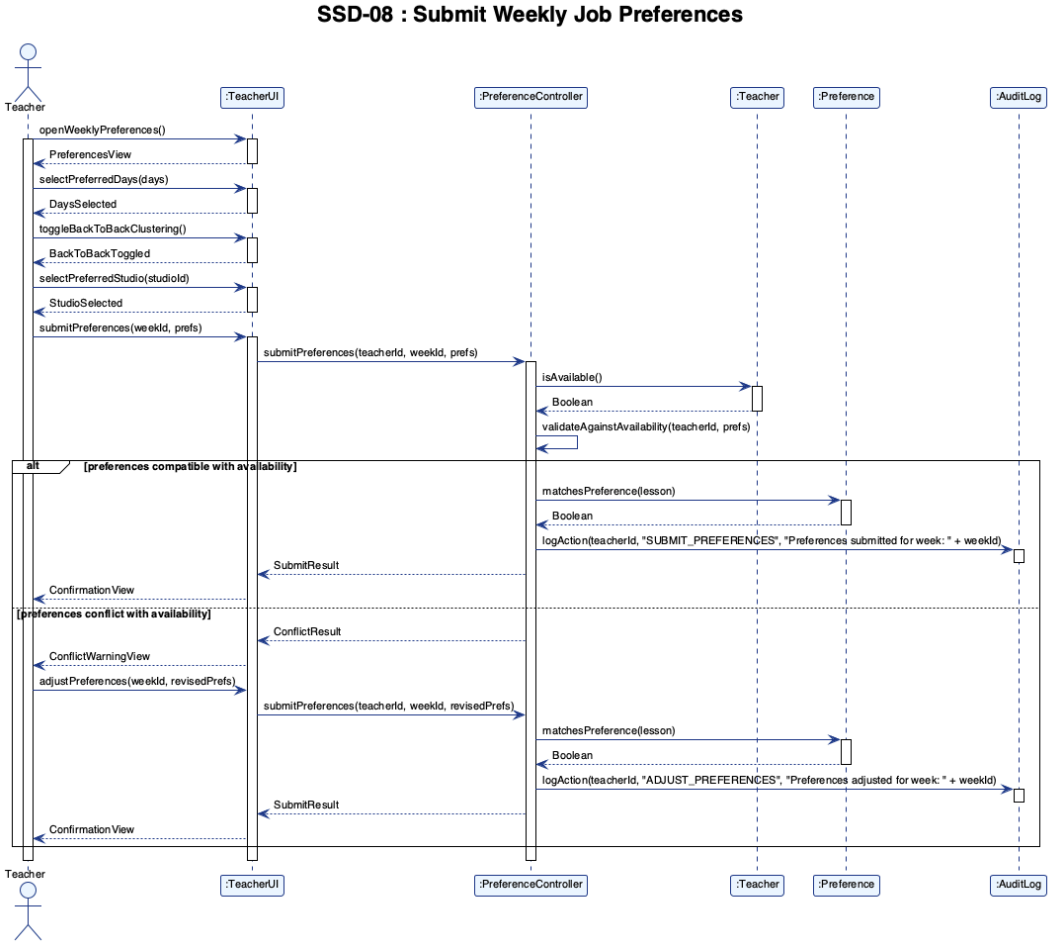


Figure 4.10: SSD-08 Submit Weekly Job Preferences

SSD-09: View Confirmed Schedule & Workload (UC-09)

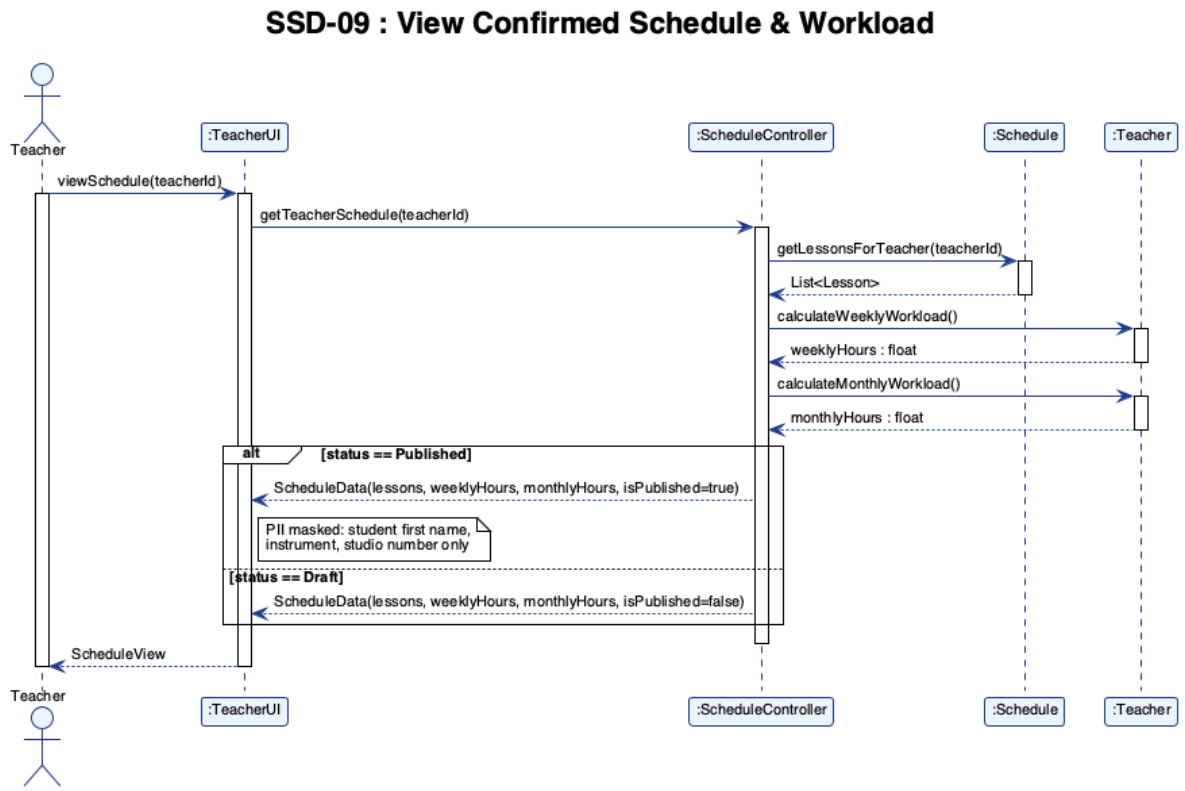


Figure 4.11: SSD-09 View Confirmed Schedule & Workload

SSD-10: Submit Job Rejection Request (UC-10)

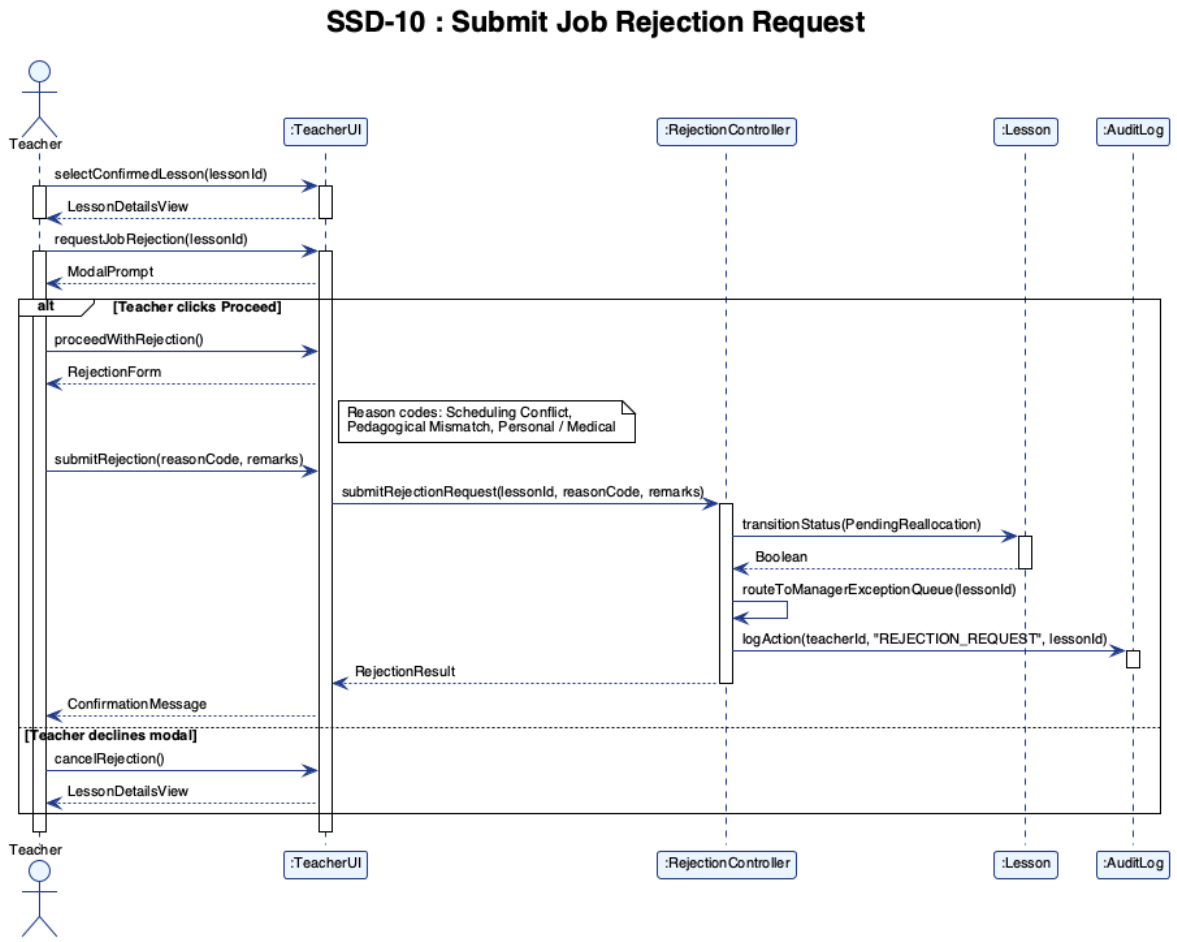


Figure 4.12: SSD-10 Submit Job Rejection Request

SSD-11: Request Peer Lesson Swap (UC-11)

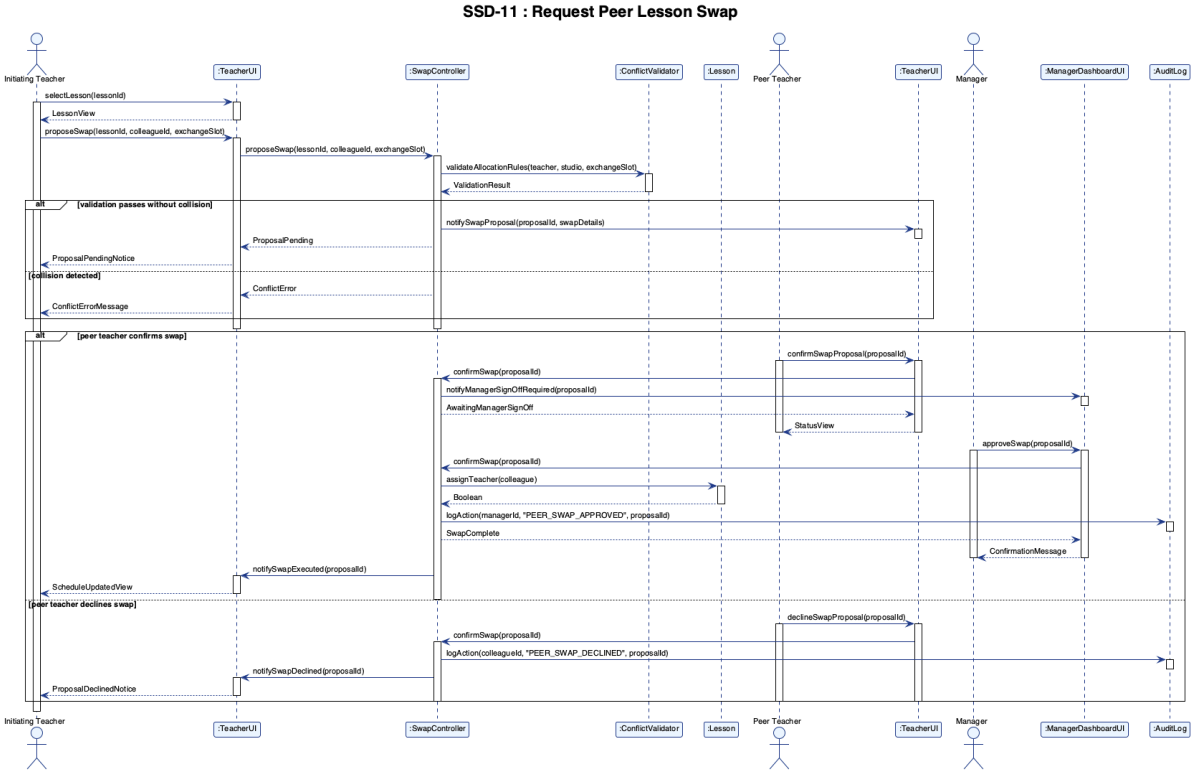


Figure 4.13: SSD-11 Request Peer Lesson Swap

SSD-12: Onboard New Staff & Manager Accounts (UC-12)

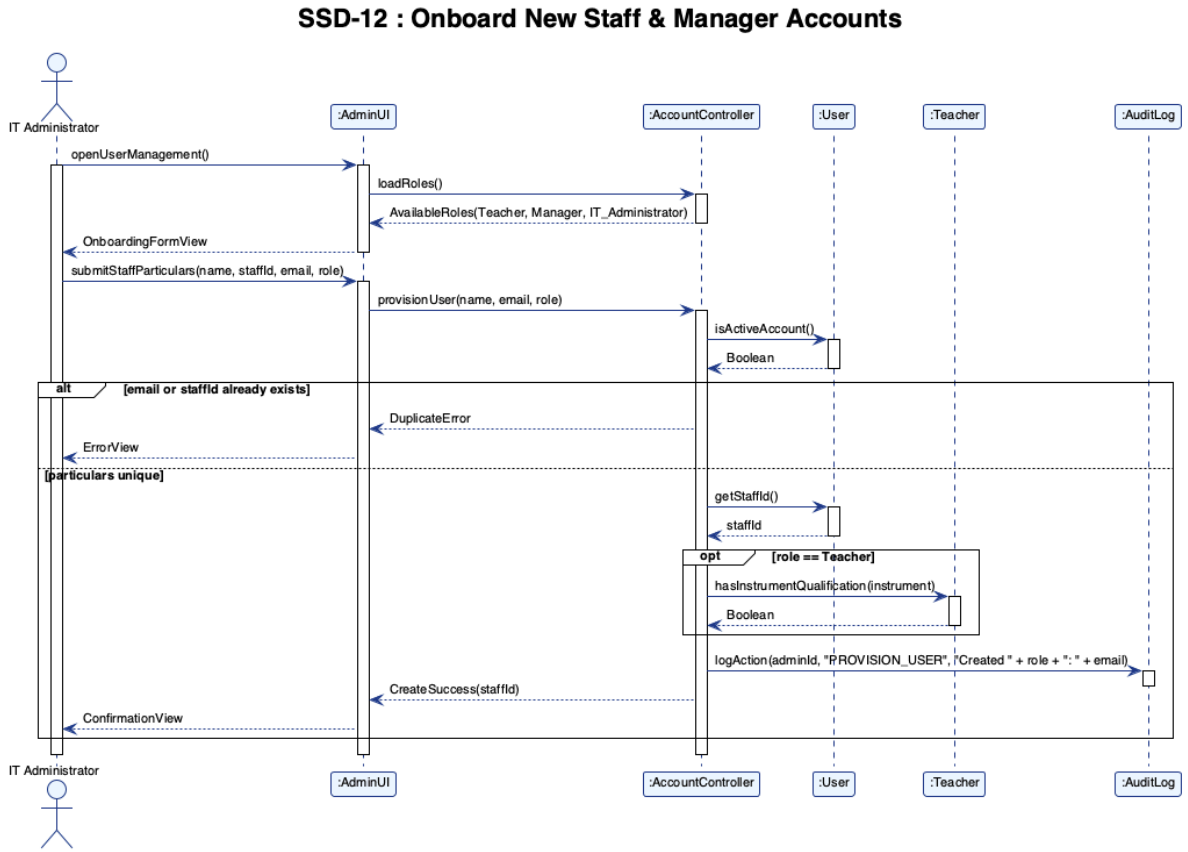


Figure 4.14: SSD-12 Onboard New Staff & Manager Accounts

SSD-13: Manage User Account Status & Access (UC-13)

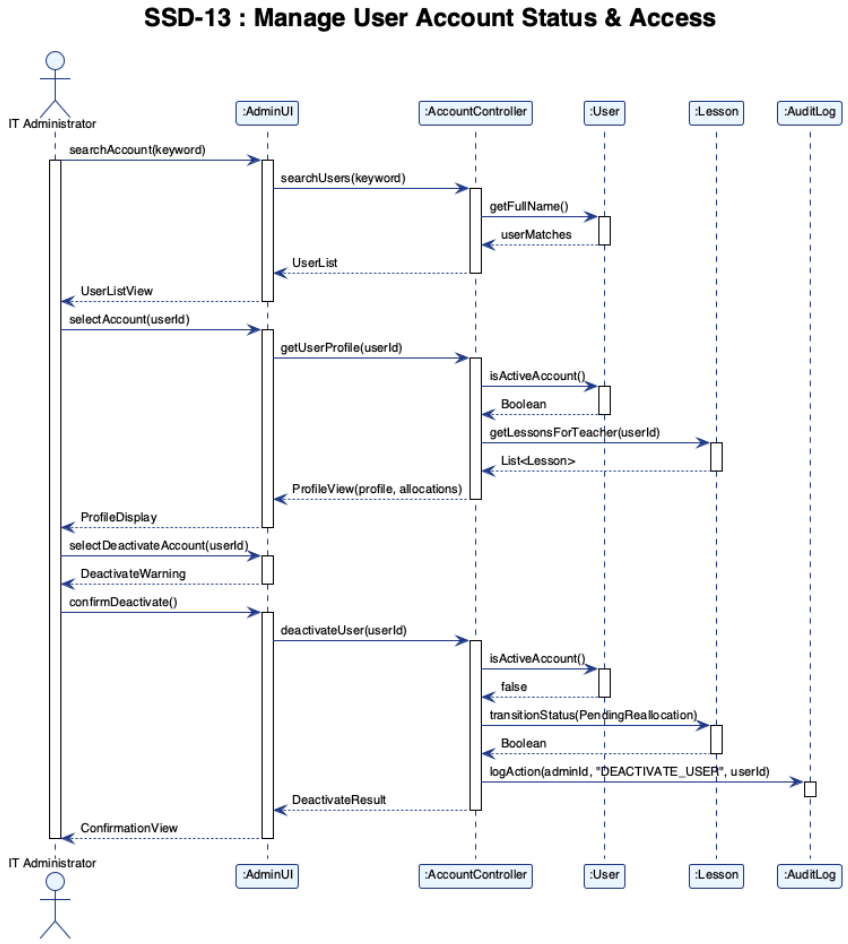


Figure 4.15: SSD-13 Manage User Account Status & Access

SSD-14: Configure Studio & Instrument Setup (UC-14)

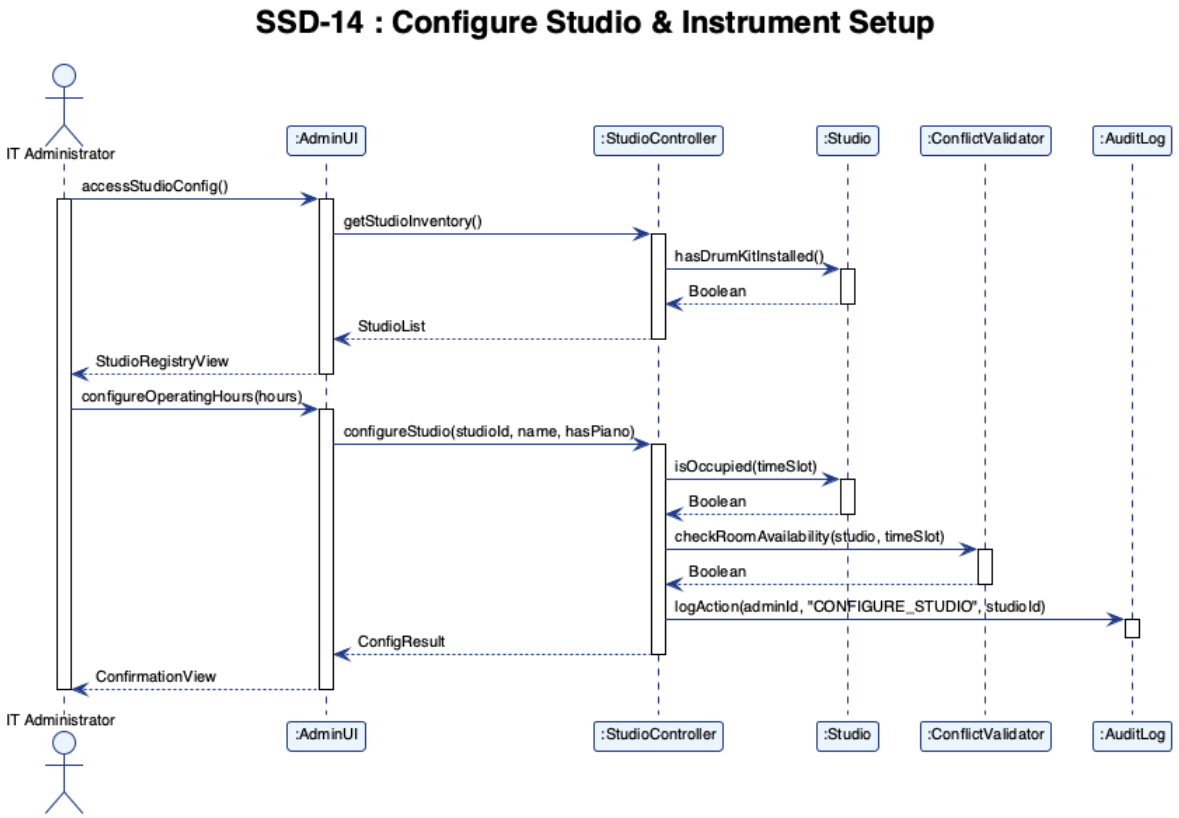


Figure 4.16: SSD-14 Configure Studio & Instrument Setup

SSD-15: Flag Facility Maintenance Window (UC-15)

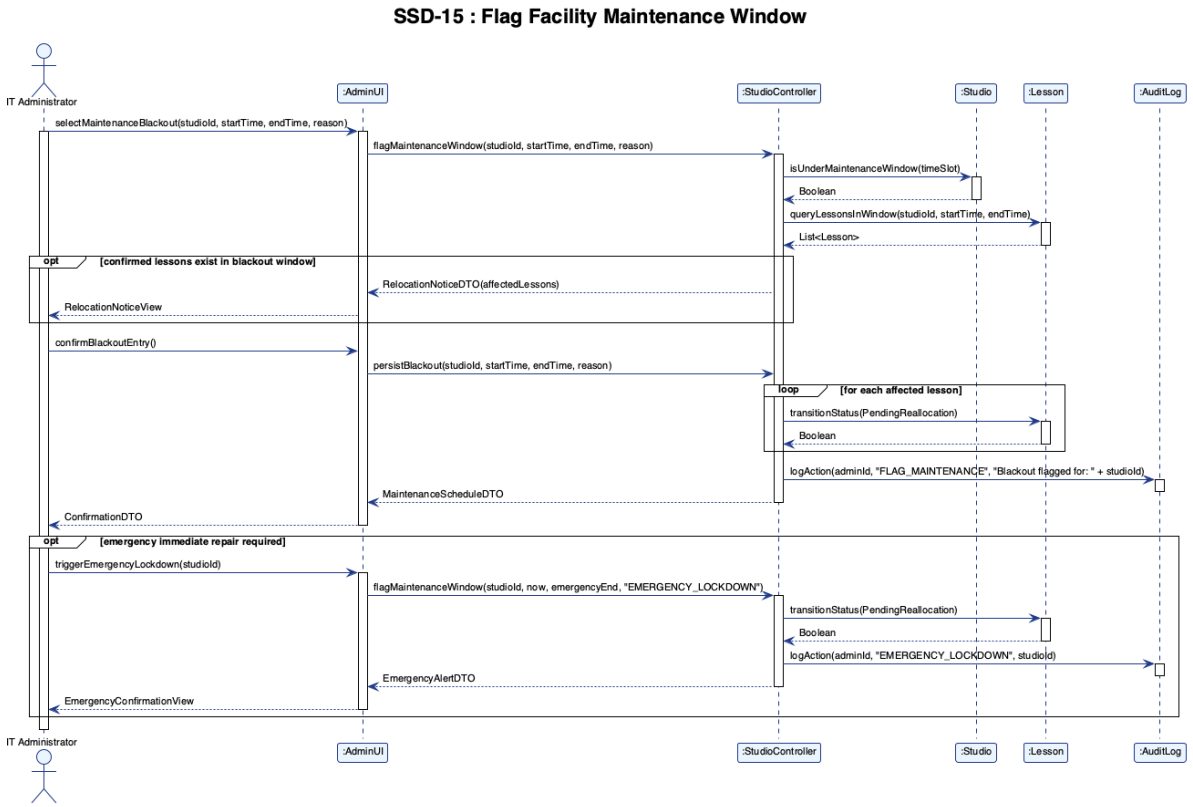


Figure 4.17: SSD-15 Flag Facility Maintenance Window

SSD-16: Generate Studio Utilization Report (UC-16)

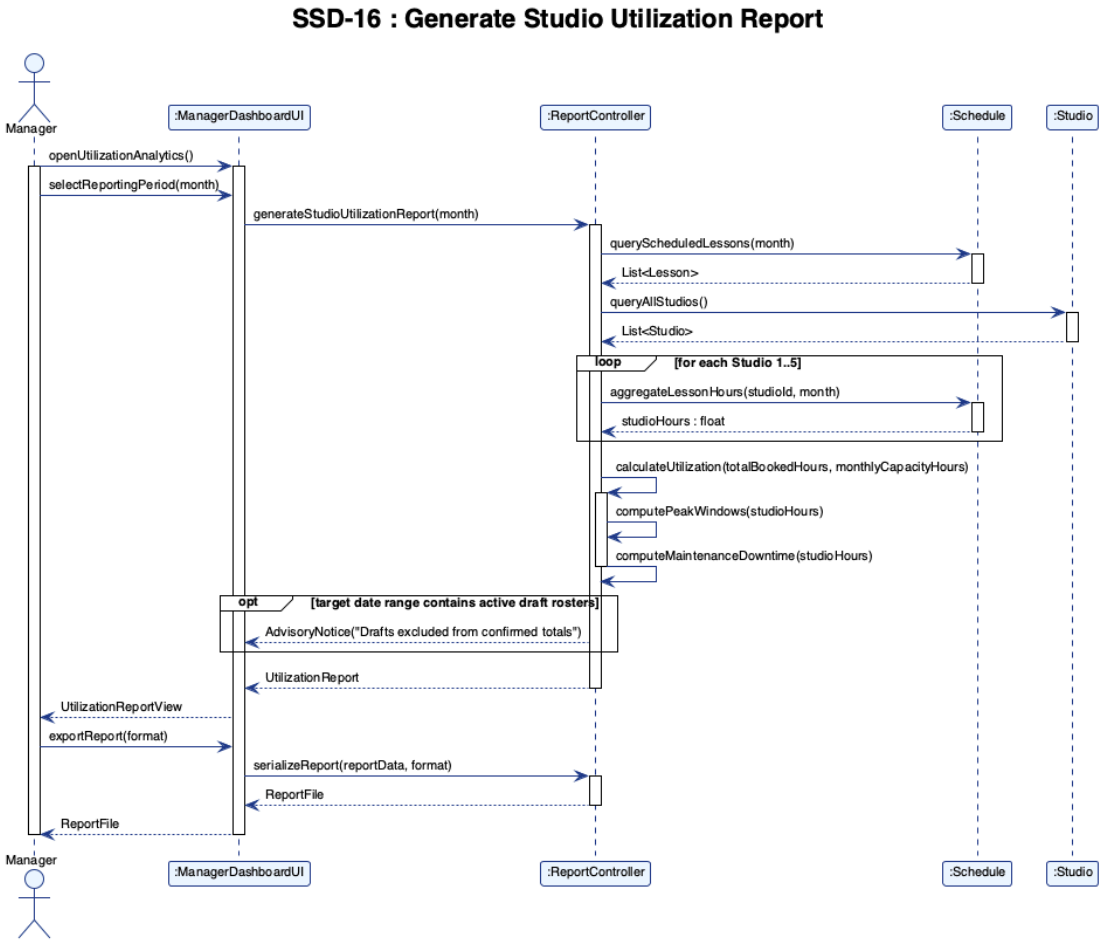


Figure 4.18: SSD-16 Generate Studio Utilization Report

SSD-17: Review Labor Compliance & Audit Trail (UC-17)

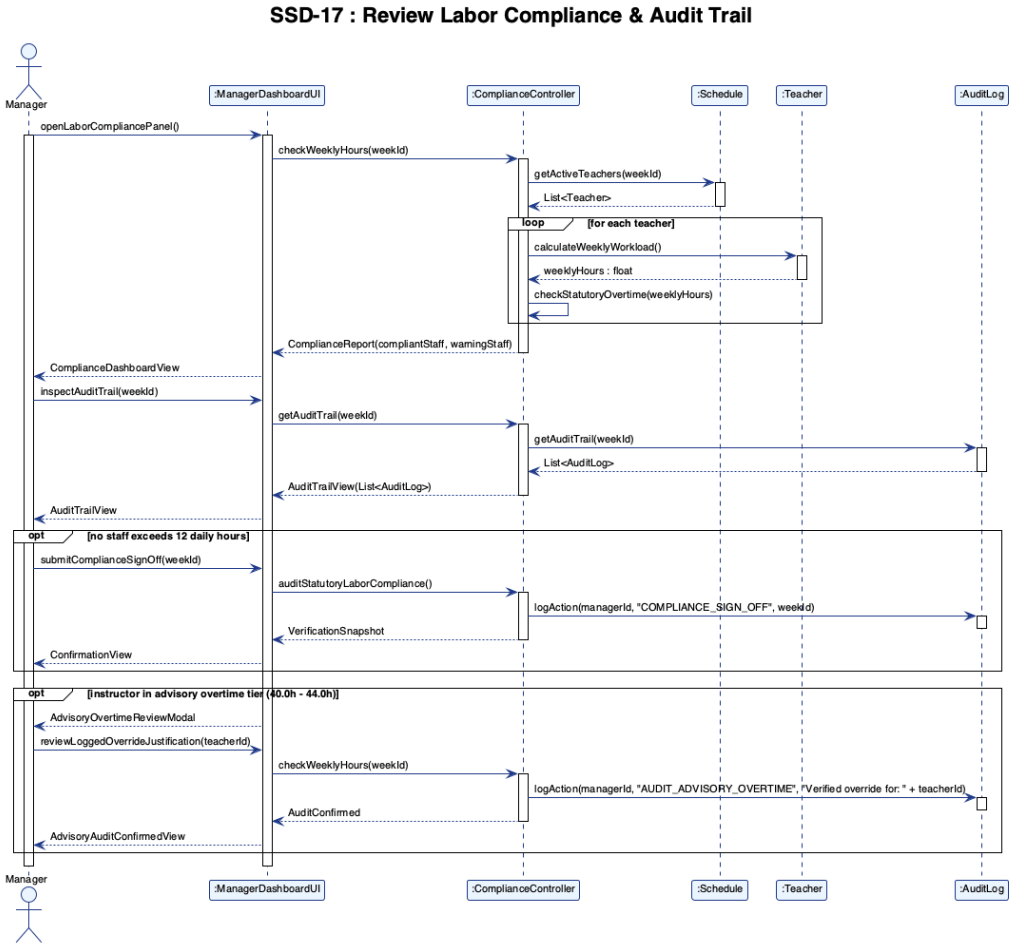


Figure 4.19: SSD-17 Review Labor Compliance & Audit Trail

5. Project Management

5.1 Work Breakdown Structure (WBS)

The WBS has three levels (Phases -> Tasks -> Subtasks). Phases 1.0 to 5.0 cover Milestone 1; phases 6.0 to 9.0 outline the Milestone 2 roadmap:

Table 5.1: 3-Level Work Breakdown Structure (WBS)

WBS Code	Task / Subtask Title	Owner	Primary Deliverable
1.0	**Project Inception & Governance (M1)**		
1.1	Project Charter & Scope Baseline	All	Project Charter
1.1.1	Problem domain analysis & client brief review	Ryan	Scope Summary
1.1.2	Team role allocation & engineering leadership roster	Joseph	Team Roster
1.2	Project Governance & Standups	Zayne	Governance Plan
1.2.1	Weekly standup facilitation & progress monitoring	Zayne	Standup Notes
1.2.2	Milestone 1 peer evaluation review	Zayne	Peer Evaluations
1.3	Stakeholder Identification & Mapping	Ryan	Stakeholder Roster
1.3.1	Operational persona definition & external engagement setup	Ryan	Engagement Plan
2.0	**Requirements Engineering (M1)**		
2.1	Stakeholder Elicitation Execution	All	Interview Minutes
2.1.1	Management & Teacher interview sessions (MIN-01, MIN-02)	Joseph	Operational Minutes
2.1.2	IT Admin, Student & Management sessions (MIN-03 to MIN-05)	Yan Qi	Governance Minutes
2.1.3	Preliminary faculty questionnaire survey synthesis	Ryan	Survey Summary
2.2	Software Requirements Specification	Ryan	SRS Document
2.2.1	Functional Requirements formulation (FR-01 to FR-19)	Ryan	FR Matrix Table 2.1

2.2.2	Quantifiable Non-Functional Requirements (14 NFRs)	Ryan	NFR Matrix Table 2.3
2.2.3	Requirements Traceability Matrix compilation	Yan Qi	RTM Table 2.4
2.3	Requirements Verification & Legal Audit	Wei Xiang	Audit Report
2.3.1	Employment Act Part IV legal compliance review	Wei Xiang	Statutory Notes
2.3.2	Cross-consistency verification across stakeholder inputs	Joseph	Verification Sign-off
3.0	**Behavioral Modeling (M1)**		
3.1	Use Case Architecture	Wei Xiang	Use Case Catalog
3.1.1	Actor boundary definition & Use Case Diagram drafting	Wei Xiang	Use Case Diagram
3.1.2	Formal specifications for 17 use cases (UC-01 to UC-17)	Wei Xiang	Use Case Specs §3.3
3.1.3	Use case review audit	Wei Xiang	Use Case Review Sign-off
3.2	Dynamic Activity Modeling	Joseph	Activity Diagrams
3.2.1	Flagship weekly job allocation workflow modeling	Joseph	Allocation Diagram Fig 3.2
3.2.2	2-stage rejection lifecycle modeling	Wei Xiang	Rejection Diagram Fig 3.3
3.2.3	Monthly availability cutoff & exception review modeling	Joseph	Cutoff Diagram Fig 3.4
4.0	**Object-Oriented Analysis (M1)**		
4.1	4-Step Object-Oriented Derivation	Joseph	OOA Spec §4.1
4.1.1	Solution abstract & candidate noun filtering matrix	Joseph	Noun Table §4.1.2
4.1.2	Class-Responsibility-Collaborator (CRC) matrix derivation	Joseph	CRC Matrix §4.1.3
4.1.3	Structural relationship derivation (generalization/composition)	Joseph	Relationship Notes §4.1.4
4.2	Domain Class Modeling	Joseph	Class Diagrams

4.2.1	Domain entity model with enums & multiplicities	Joseph	Entity Diagram Fig 4.1
4.2.2	Boundary-Control ECB architectural dependency map	Joseph	Control Diagram Fig 4.2
4.3	Sequence Diagram Authoring	All	17 Sequence Diagrams
4.3.1	Manager sequence diagrams (SSD-01 to SSD-06)	Wei Xiang	SSD-01 to 06
4.3.2	Teacher sequence diagrams (SSD-07 to SSD-11)	Tan Yan Qi	SSD-07 to 11
4.3.3	IT Admin & Compliance diagrams (SSD-12 to SSD-17)	Zayne Lee	SSD-12 to 17
4.4	OOA Consistency Audit	Tan Yan Qi	OOA Audit Sign-off
5.0	**Milestone 1 Integration & Submission**		
5.1	Report Assembly & Formatting Audit	Ryan	M1 Report Draft
5.1.1	Cross-reference consistency check	Ryan	Cross-reference Checklist
5.2	Visual Diagram Rendering & PDF Generation	Yan Qi	Submission PDF
5.3	Oral Defense Rehearsal & Slide Deck Prep	Wei Xiang	Defense Deck
6.0	**Architecture & High-Level Design (M2 Roadmap)**		
6.1	Architectural pattern selection (MVC/ECB) & component diagrams	Joseph	Architecture Spec
6.2	Design pattern implementation (Strategy for tie-breaking)	Wei Xiang	Design Pattern Spec
7.0	**Interactive Web Prototype (M2 Roadmap)**		
7.1	UI/UX responsive wireframes (desktop matrix & mobile grid)	Zayne	UI Wireframes
7.2	Frontend & backend decision-support prototype build	Joseph	Working Prototype
8.0	**Verification & Validation (M2 Roadmap)**		

8.1	Black-box test suite (equivalence partitioning & decision tables)	Wei Xiang	Black-Box Test Cases
8.2	White-box test suite (basis path testing & cyclomatic complexity)	Yan Qi	White-Box Test Cases
9.0	**Milestone 2 Closure & Final Demonstration**		
9.1	Final project report compilation & technical demo handoff	Joseph	M2 Final Report

5.2 Project Timeline & Critical Path Analysis

The project schedule was executed between Monday, 31 August 2026, and Friday, 9 October 2026, culminating in Milestone 1 report submission:

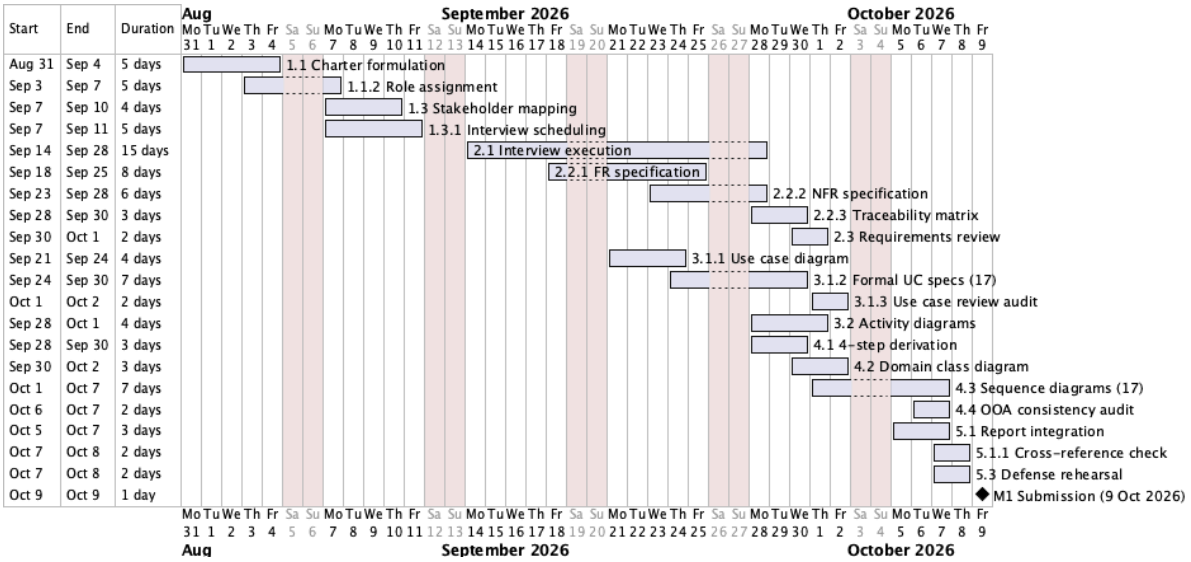


Figure 5.1: Milestone 1 Project Schedule & Timeline (Gantt Chart)

Critical Path Analysis

The critical path is:

> **1.1 Charter Formulation → 1.3 Stakeholder Mapping → 1.3.1 Interview Scheduling → 2.1 Interview Execution → 2.2.1 FR Specification → 3.1.1 Use Case Diagram → 3.1.2 Formal UC Specifications → 4.1 4-Step Derivation → 4.2 Domain Class Diagram → 4.3 Sequence Diagrams → 5.1 Report Integration → 5.1.1 Cross-Reference Check → M1 Submission (9 Oct 2026)**

Total critical path duration spans exactly 30 working days across 6 academic weeks. Non-critical parallel paths—such as NFR specification (2.2.2), Traceability Matrix compilation (2.2.3), Activity diagram authoring (3.2), and Oral Defense rehearsal (5.3)—contain buffer margins allowing task overlap without impacting the final 9 October 2026 submission deadline.

6. References

- Guest, D. E. (2002) 'Perspectives on the Study of Work-life Balance', *Social Science Information*, 41(2), pp. 255-279. doi: 10.1177/0539018402041002005.
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7. Appendix A: Stakeholder Elicitation Minutes

Formal meeting minutes for all five stakeholder engagement sessions are presented below.

A.1 MIN-01-MGR: Management Operations (Marcus Tan)

Date: 15 September 2026 | **Duration:** 75 min | **Venue:** Music School Admin Office / MS Teams
Interviewee: Marcus Tan (Operations Co-Manager)
Attendees: Joseph Poon (Lead Interviewer), Ryan Ke (Minute Taker), Marcus Tan

Objectives: Elicit operational workflows, scheduling heuristics, and pain points for weekly allocations; establish business rules for workload caps, rest periods, and overtime thresholds; formalize late availability and job rejection exception handling.

Key Findings:

- **Manual Overhead:** Constructing weekly rosters in unversioned Excel spreadsheets consumes an estimated 8 to 12 hours per cycle and frequently results in undetected room double-bookings.
- **3-Tier Workload Badges:** Workload indicator requires three tiers: Green (<35.0h), Amber (35.0h–40.0h), and Red (>40.0h). The top 3 lowest-workload teachers must be displayed prominently to guide allocation equity.
- **Side-by-Side Allocation Matrix:** Manager compares up to 3 candidate teachers side-by-side, reviewing availability, running hours, preferences, and physical studio location for adjacent slots on that date.
- **Business Constraints:** Instrument qualification is a hard constraint; continuous teaching is capped at 4.0h with a mandatory 1.0h rest break; Studio 5 exclusively houses the drum kit.
- **Overtime Policy:** 40.0h threshold triggers an advisory soft warning (40.5h–44.0h tier) that managers can override with logged justification; 44.0h statutory ceiling is enforced as an unbreachable hard block under the Employment Act.
- **Exception Queues:** Late availability changes (post-19th) and teacher job rejections route to dedicated manager exception queues for review.
- **Audit Logging:** Immutable audit trail must log all allocations, overrides, and reassignments, retained for >= 24 calendar months.

A.2 MIN-02-TCH: Teacher Operations (Elena Lim)

Date: 17 September 2026 | **Duration:** 60 min | **Venue:** Faculty Lounge / MS Teams
Interviewee: Elena Lim (Senior Piano & Violin Instructor)
Attendees: Ryan Ke (Lead Interviewer), Wei Xiang (Minute Taker), Elena Lim

Objectives: Understand teacher availability submission lifecycle; elicit preferences regarding lesson clustering and room acoustics; establish job rejection protocols; define mobile usability requirements.

Key Findings:

- **Availability Horizon & Cutoff:** Teachers plan commitments up to 1 month ahead. Availability grid must remain open for additions and edits until 23:59 on the 19th of each month, after which changes require manager review.
- **Job Preferences:** Instructors strongly prefer back-to-back lesson clustering to eliminate unpaid idle gaps, alongside preferred teaching days and studio acoustic selection.
- **Fatigue Protection:** 4.0-hour continuous teaching ceiling strongly endorsed; instructors require at least 1.0 hour of rest before resuming instruction.
- **Schedule Transparency:** Teachers currently maintain private shadow logs due to spreadsheet inaccuracies; system must provide a personal weekly schedule with running monthly hours.
- **2-Stage Rejection Protocol:** Rejection requests must present a mandatory warning modal instructing the teacher to discuss the change with the manager, followed by standardized reason codes (`Scheduling Conflict`, `Pedagogical Mismatch`, `Personal / Medical`).
- **Mobile Usability:** Smartphone is the primary device between lessons; interface must feature touch targets >= 44x44pt, zero horizontal scrolling, and local draft caching for network resilience.

A.3 MIN-03-IT: IT Infrastructure & Administration (Alex Wong)

Date: 22 September 2026 | **Duration:** 60 min | **Venue:** Campus IT Laboratory / In-Person
Interviewee: Alex Wong (Systems Administrator)
Attendees: Tan Yan Qi (Lead Interviewer), Zayne Lee (Minute Taker), Alex Wong

Objectives: Define user onboarding and deactivation workflows; establish role-based access control boundaries; model physical studio facilities and maintenance blackout windows; verify statutory data protection.

Key Findings:

- **Account Governance:** IT admins provision accounts with Staff ID, name, email, contact, and instrument qualifications. Physical record deletion is prohibited; soft deactivations preserve historical auditability.
- **Role-Based Isolation:** Managers hold full operational access; Teachers access only personal timetables; IT Administrators configure facilities and users but cannot modify teaching allocations.
- **Facility Modeling:** Academy houses 5 studios; Studios 1–4 house pianos; Studio 5 houses a piano and acoustic drum kit with soundproofing; drum lessons must be restricted to Studio 5.
- **Maintenance Windows:** Administrators can schedule servicing windows (eg piano tuning), during which the room is suppressed from the allocation matrix and conflicting lessons are relocated.
- **Statutory Data Protection:** Personal Data Protection Act (PDPA Section 24) requires encryption at rest/transit and masking of student contact details on non-administrative screens.

- **Audit Retention:** Immutable, append-only audit trail retained for >= 24 calendar months for regulatory inspection.

A.4 MIN-04-STU: Student Experience & Scheduling (Chloe Teo)

Date: 24 September 2026 | **Duration:** 45 min | **Venue:** Academy Student Waiting Lounge / In-Person
Interviewee: Chloe Teo (Student Council Representative)
Attendees: Zayne Lee (Lead Interviewer), Tan Yan Qi (Minute Taker), Chloe Teo

Objectives: Assess student scheduling friction, lesson reschedule communication, and studio constraints; define expectations for student continuity.

Key Findings:

- **Lesson Structure:** Students enroll in recurring weekly 30-minute individual sessions, with peak demand during weekday evenings (17:00–21:00) and weekends (08:00–21:00).
- **Rescheduling Delays:** Rescheduling via email or phone currently takes 24 to 48 hours; students require rapid turnaround and prioritized rescheduling when instructor rejections occur.
- **Teacher Continuity:** Students prioritize consistent instruction with their primary assigned teacher; substitutions should only occur during emergencies.
- **Studio 5 Contention:** Drum students face scheduling bottlenecks due to the single drum room; piano, violin, and trumpet students enjoy greater room flexibility.
- **System Boundary Confirmation:** Students do not log into the Milestone 1 system; all lesson booking and adjustments are managed by academy staff on their behalf.

A.5 MIN-05-MGT-EXT: School Governance & Regulatory Compliance (David Koh)

Date: 28 September 2026 | **Duration:** 75 min | **Venue:** Executive Conference Room / MS Teams
Interviewee: David Koh (Academy Director & Compliance Officer)
Attendees: Wei Xiang (Lead Interviewer), Joseph Poon (Minute Taker), David Koh

Objectives: Formalize strategic operational metrics and studio utilization targets; establish institutional hierarchy balancing occupancy against teacher retention; define compliance under Employment Act Part IV.

Key Findings:

- **Perishable Capacity & Profitability:** Every unbooked 30-minute slot during operating hours represents permanently lost revenue; system must track Room Deficit vs. Teacher Deficit.

- **Governance Priority Hierarchy:** Strategic priorities are ranked: (1) Statutory legal compliance, (2) Student-teacher continuity, (3) Teacher retention and workload equity, (4) Studio occupancy. Retention and work-life balance are explicitly prioritized over filling vacant rooms.
- **Retention Alerts:** System must flag instructors receiving <50% of their requested monthly target hours to prevent faculty turnover.
- **Statutory Labor Compliance:** Singapore's Employment Act 1968 (Part IV Section 38) governs standard weekly hours (44h normal cap), statutory overtime pay (1.5x basic hourly rate), and a 12-hour maximum daily work spread.
- **Audit Verification:** Academy requires an append-only audit trail logging all allocation transactions, late exceptions, overrides, and deactivations, retained for >= 24 calendar months.
- **Facility Servicing:** Maintenance blackout windows must be integrated into operational scheduling to prevent room booking collisions during instrument tuning.